

Video Astronomie with Smart Telescopes?

Björn Kattentidt

ESOP 45

Toulouse, France

22./23. Aug. 2026



Seestar S50

Observation of occultation of
(5368) Vitagliano
March 12, 2026 with Bernd Gährken
north of Regensburg

- Video with Seestar app
- Timestamp?? 1pps flasher
- Occult Watcher filter
- Signal-to-noise ratio?
- Magnitudes of S50 in video
- Event filtering with Excel
- DAMIT / ISAM
- Expedition
- Results

International Occultation Timing Association European Section

2026 May 04 ~20:17 UT: (136108) Haumea occults UCAC4 524-056397 (14.7 mag)

A very important occultation, a TNO with rings and two moons!!!

<<< Have a look here >>>

Jupiter Trojan (617) Patroclus / Menoetius occults an 11.9 Vmag star across Europe on 2026 Jan 4st

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Phemu Events for Saturn, 2025-2026

Only a few events for the period 2025 to 2026 are observable with a reasonable chance of success. More infos you can find here.

Occultation by the Jovian satellite Ganymede in the early morning of the 14th of October 2025 - A European campaign

More infos and a meeting you can find here.

Observation report

Successful Observation of TNO 2013 LU28 Stellar Occultation in the far

Newest JOA ready for download!



Report for the Seestar telescope written by Christian Weber

This article describes aspects of the use of the

At the moment we don't support "**X**" anymore.

Here you can find us on **Mastodon**

The XLV ESOP meeting will be held in Toulouse (France), from 22 till 23 August 2026, with extra days for excursions. All the occultation community is cordially invited.

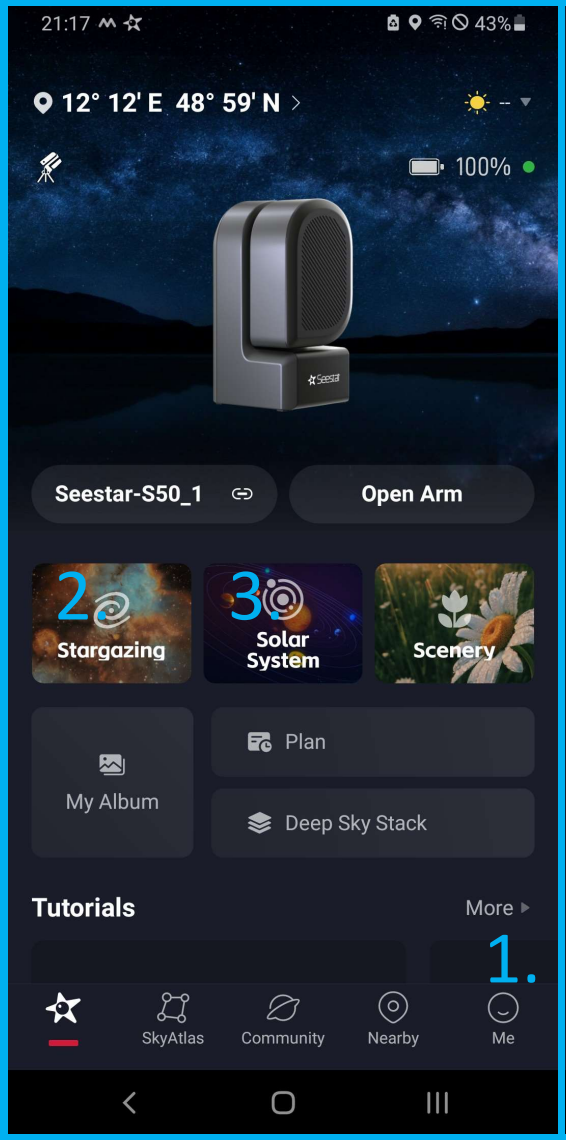
The Journal for Occultation Astronomy

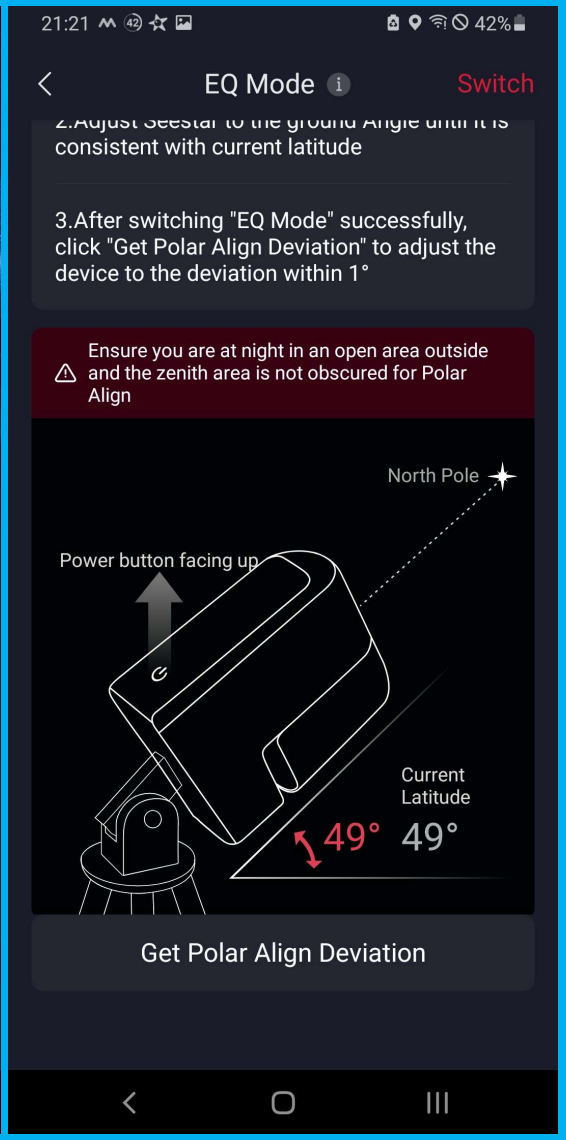
is now free for all!

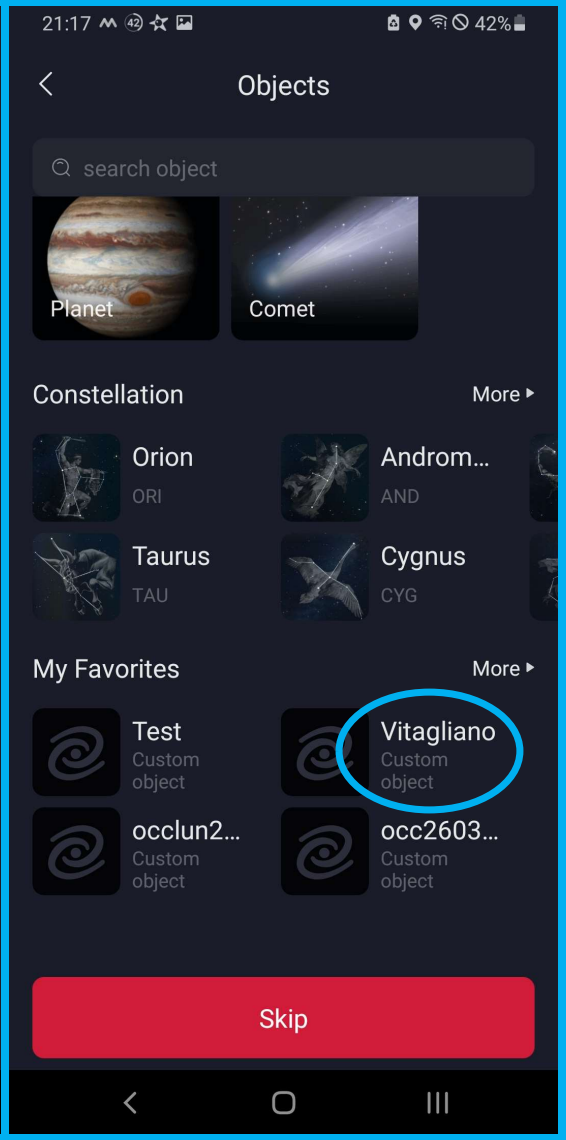
Call for Observations
Occultations of Stars by Minor Planets

Call for Observations

ursus
Internetzugriff

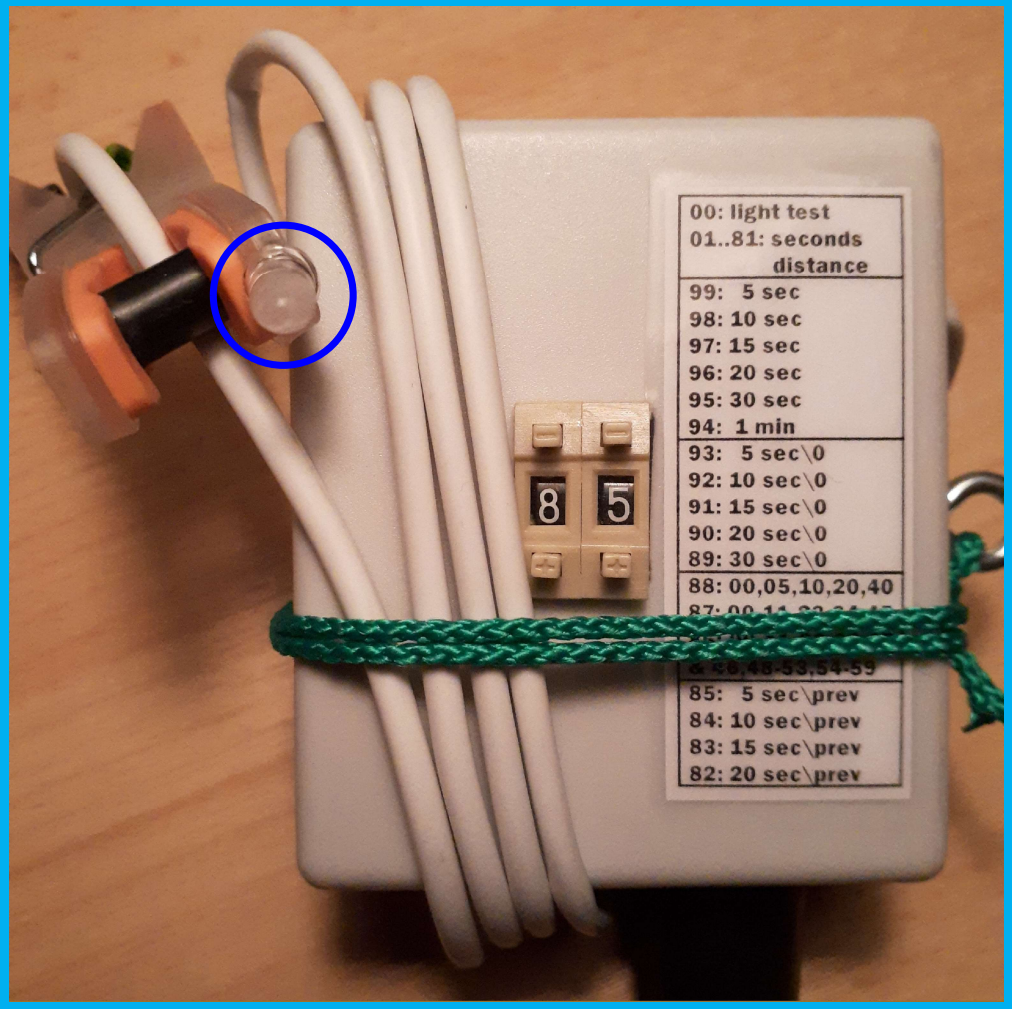

21:17
43%
12° 12' E 48° 59' N
100%
Seestar-S50_1
Open Arm
2
Stargazing
3
Solar System
Scenery
My Album
Plan
Deep Sky Stack
Tutorials
More
1.
SkyAtlas
Community
Nearby
Me


21:21
42%
EQ Mode
Switch
consistent with current latitude
3.After switching "EQ Mode" successfully, click "Get Polar Align Deviation" to adjust the device to the deviation within 1°
Ensure you are at night in an open area outside and the zenith area is not obscured for Polar Align
North Pole
Power button facing up
Current Latitude
49°
49°
Get Polar Align Deviation


21:17
42%
Objects
search object
Planet
Comet
Constellation
More
Orion
ORI
Androm...
AND
Taurus
TAU
Cygnus
CYG
My Favorites
More
Test
Custom object
Vitagliano
Custom object
occlun2...
Custom object
occ2603...
Custom object
Skip


12:16
96%
Neptune
98%
Slow
Fast
Track
4. Video und RAW!
=> AVI!
Raw
1x
2x
4x
Photo
Video

Timestamp? ->> 1 peak per second flash unit ->> task solved!



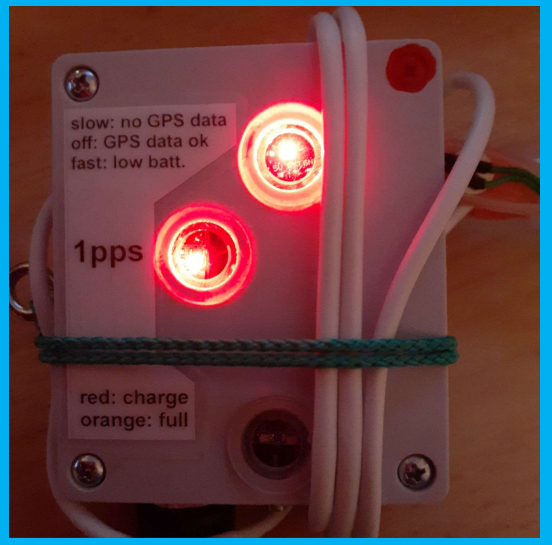
optical! – Less than 0,1ms delay

Compatible with any video system!!

40ms flash, many flash patterns

5 hours of operation

#85: 40ms flash every 5 sec except 55th



On GITHUB:

- Build Instructions
- User Manual
- Firmware Code (Arduino nano)
- Hardware PCB (EAGLE)

DIY! ->> https://github.com/leontodon1/1pps_flasher

Timestamp on the S50

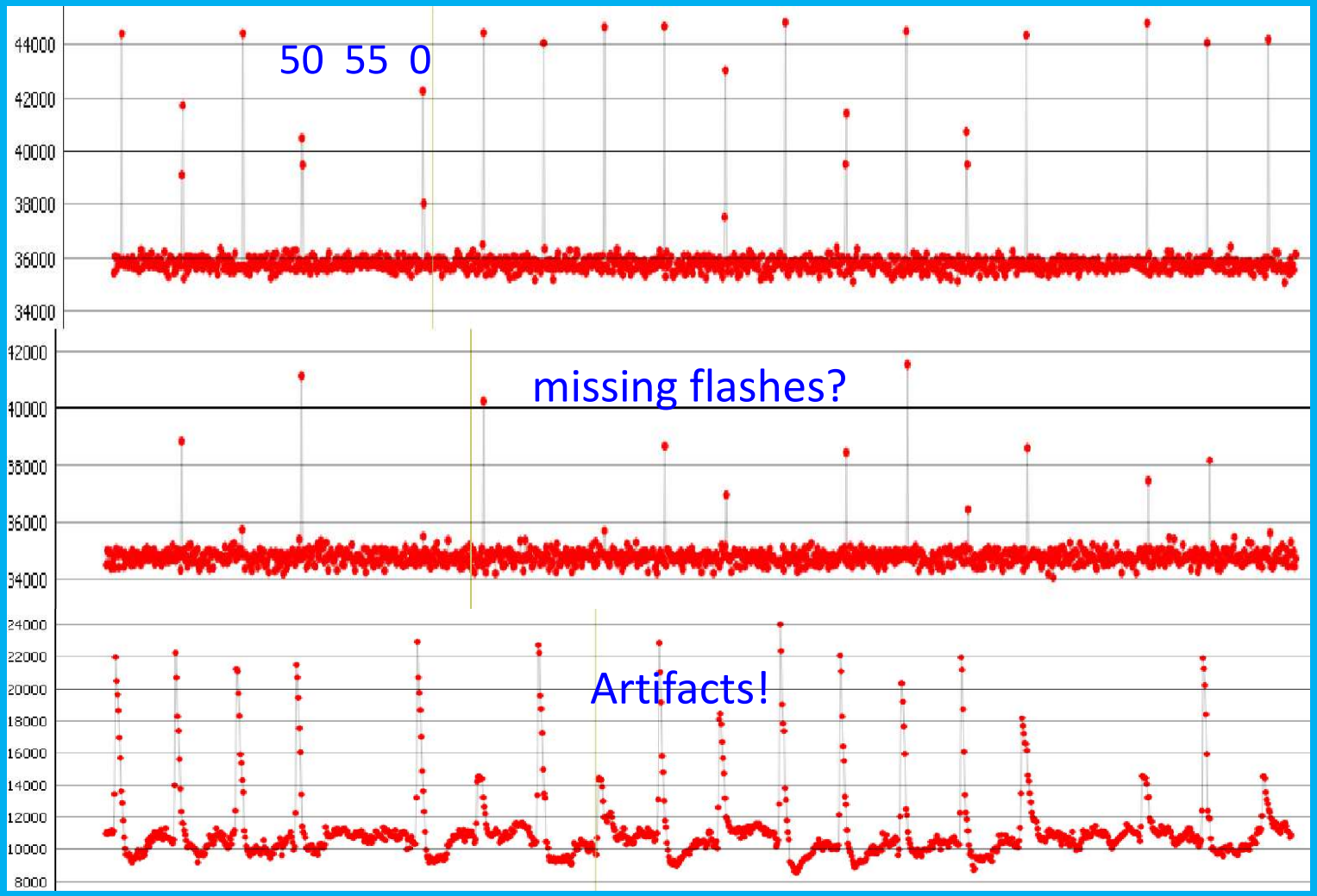
avi vs. mp4 (1x, #85)

avi „RAW“ 80ms

avi 35ms (1x)

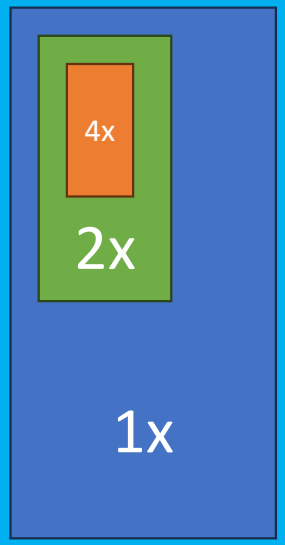
mp4 35ms

PyMovie / PyOTE

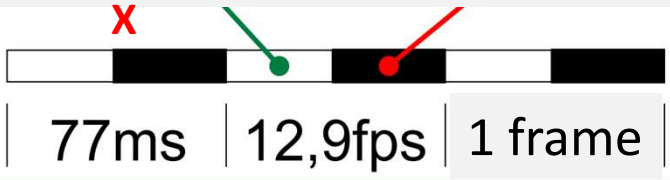


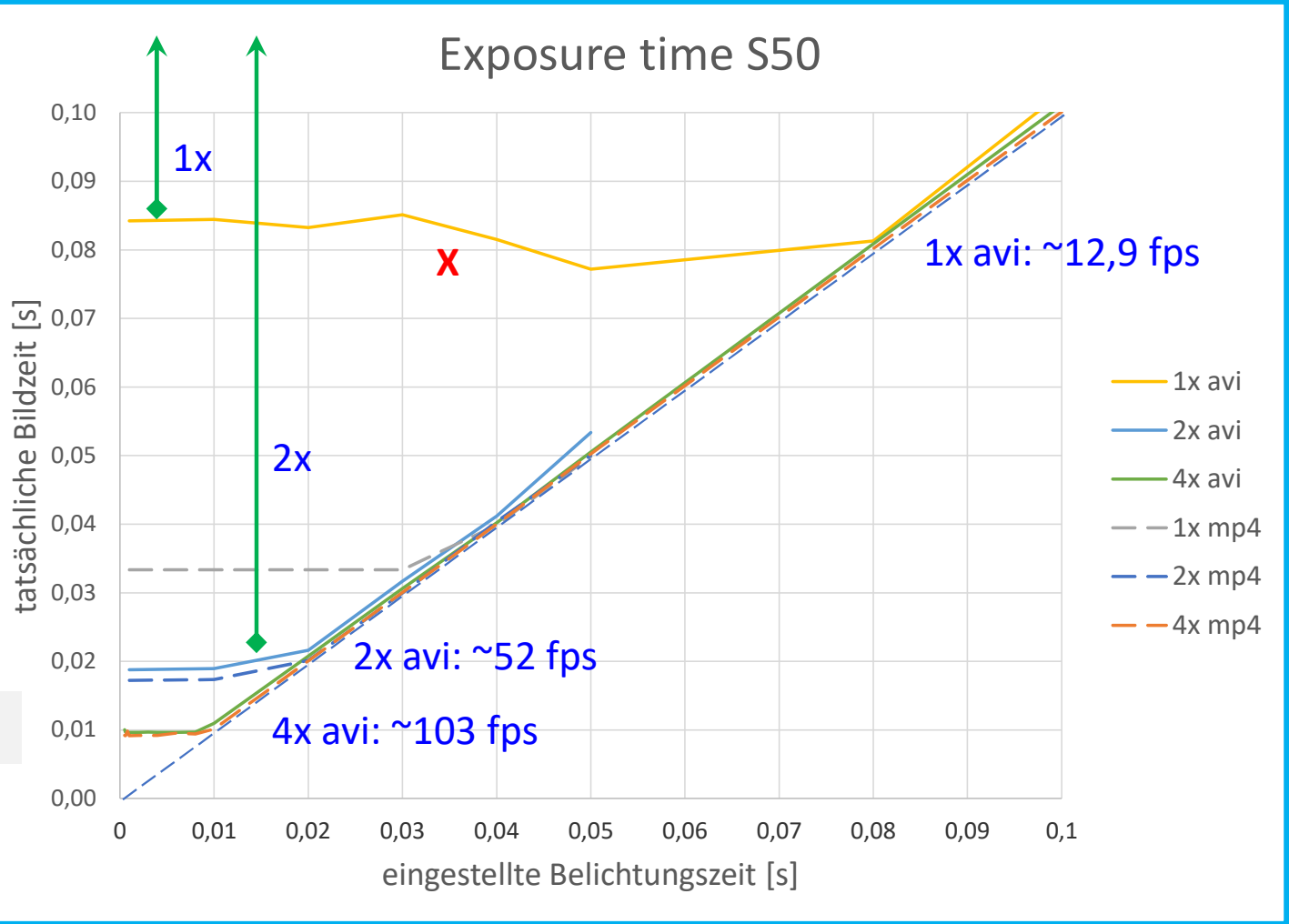
Trap! Exposure time and fps on the S50

FOV formats

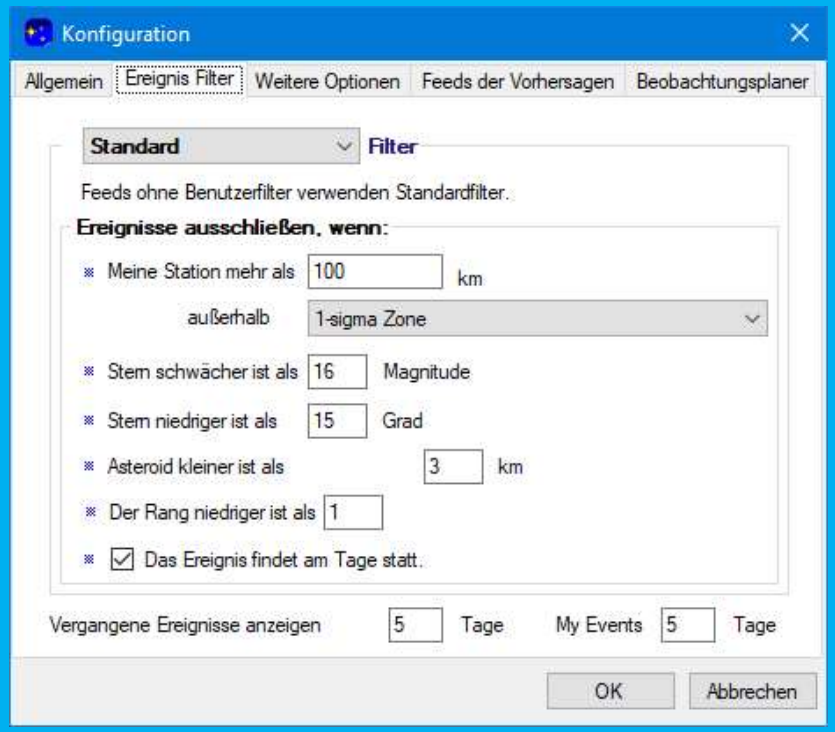


exposed 35ms, unexposed 45ms





Occult Watcher – Event Filter



Filters required for the S50 mobile:

distance earth's surface < 30 km
drop > 2,5mag

Desired result:

- max frames possible (frm_max ≥ 3)
  => Filter!
- shortest possible exposure time t_exp_min (SNR > 3)
  Martin Gutekunst, 2025 ESOP 44 at Posen/PL

Parameter for t_exp_min, frm_max

optics: focal length, aperture
camera: pixel size, binning, sensitivity
OW: duration, brightness, mag drop, distance

max. magnitude on S50 - h~45°

magnitude limit

detection limit:

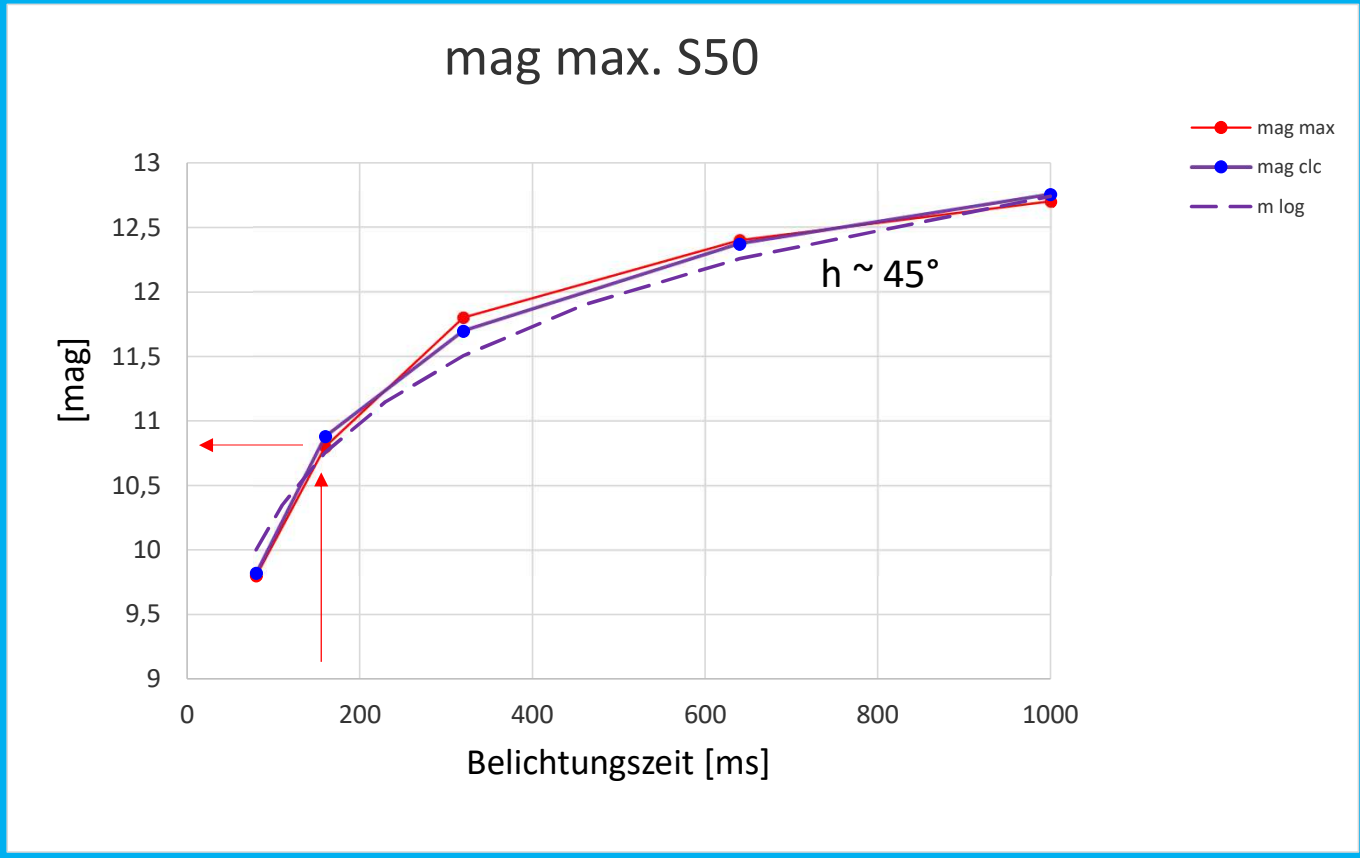
SNR << 3

determined experimentally

$$m_{log} = 10 + \frac{ln\left({t_{\exp(ms)}}/{80}\right)}{ln(100^{\,0,2})}$$

empirical formula (h ~ 45°):

.. based solely on exposure time



$$m_{\max} = (-20500 + 6800 * ln(t_{\exp(ms)}))^{0,25}$$

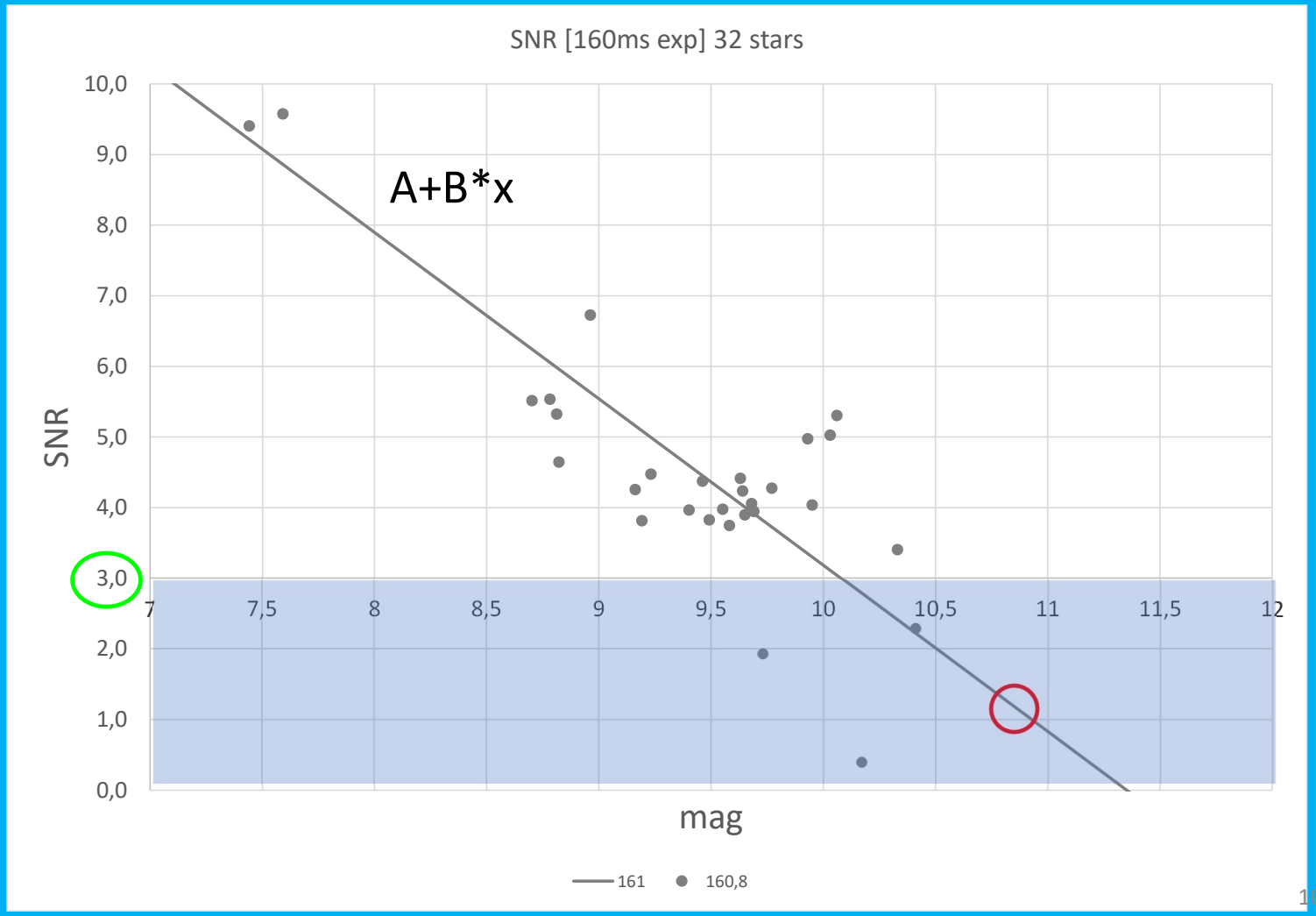
$form$: $(A + B * ln(t))^{0,25}$ empirical

Signal to noise ratio on S50

160ms video

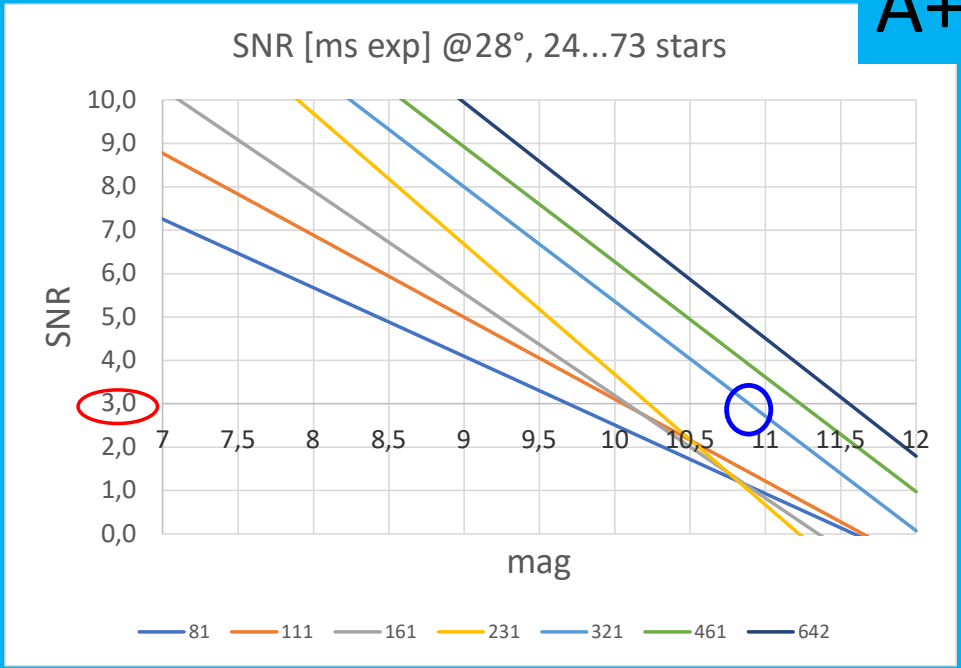
approx. 400 frames

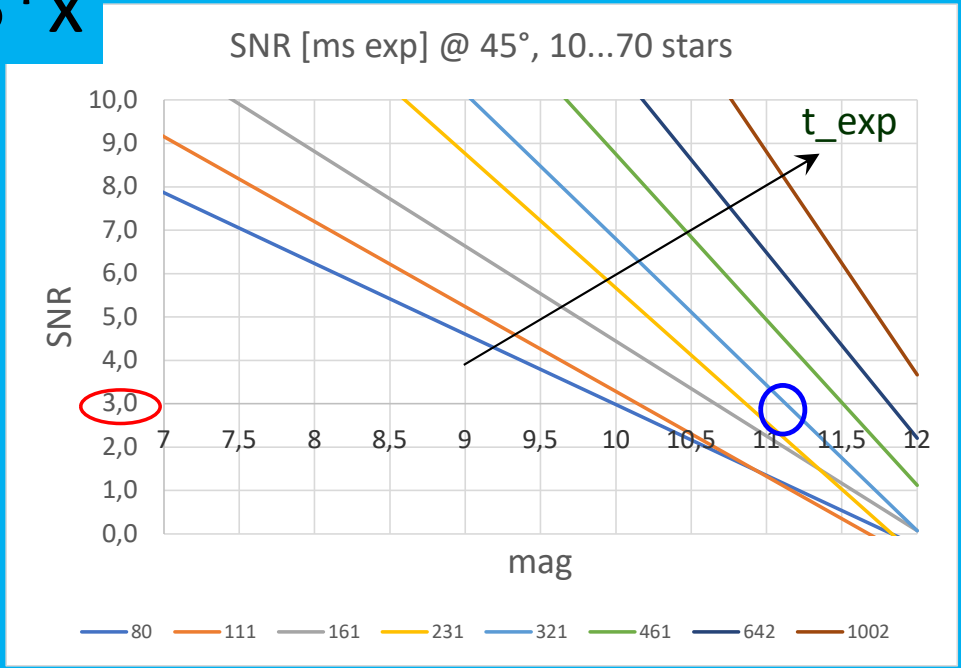
light curves of 32 stars



Signal to noise ratio on S50

A+B*x





Extinction at 28° versus 45°: approx. 0.3 mag

Extinction at 45°: ~ 0,1 mag

Magnitude on S50 @ SNR=3 & Extinction

$m_{extinction} = k * A * h^B$

$k = 0{,}2\ vis.\ (0{,}15\ cam.)$

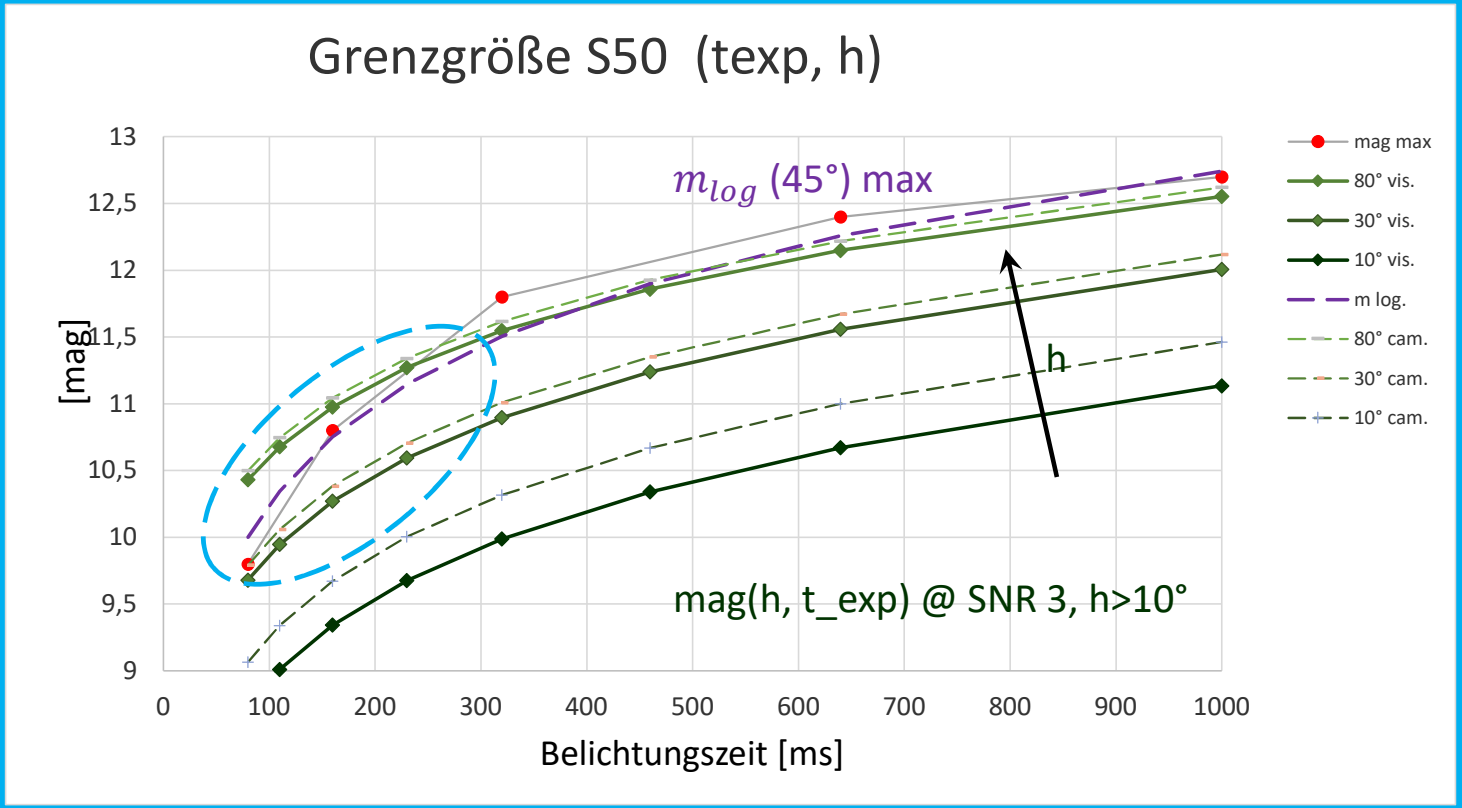
$A = 38{,}6 \quad B = -0{,}85$

$m_{log}(t_exp)$

Evaluatable only above

SNR = 3 !

exposure time, star altitude



h > 10°: $mag(t_{exp}, h) = A(h) * t_{exp}^{B(h)} - extinction$

$$mag(S50, @SNR = 3) = (6{,}6 + 0{,}014 * h) * t_{\exp(ms)}^{(0{,}088 - 0{,}0002 * h)} - m_{extinction}$$

$$mag(S50, @SNR = 3) = (7{,}9 - 0{,}014 * (90 - h)) * t_{\exp(ms)}^{(0{,}07 + 0{,}0002 * (90 - h))} - (0{,}2 + 0{,}0035 * e^{0{,}07 * (90 - h)})$$

Occult Watcher extract - 3 winter months: 2800 events

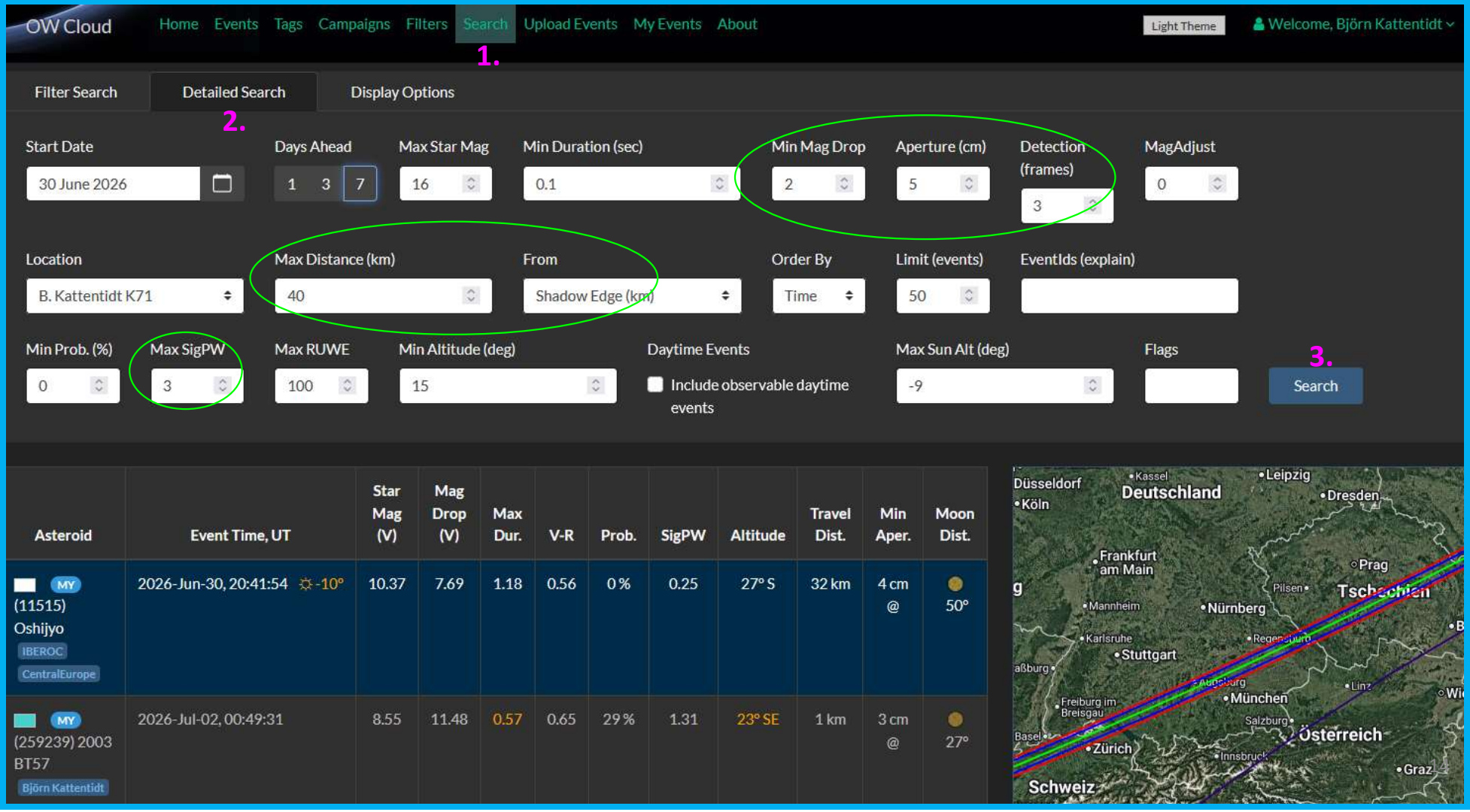
< 10mag, 5km size, 30km distance: 23 events (displayed)

< 11mag, 40km distance: 39 events

OW Filter:
<100 km dist.
<16 mag
>15 ° altitude
>3km diameter

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	30	100	2	2,5	-20500	2	160	Macro: ctrl+m only OBS	readme!	0,00%		10		S50			2,5	5	deaktivieren	
2	20	A5-B5/2	3		6800	3	320	Macro: ctrl+e ALL	time sort: AM	5,00%		15,5		C11			0,1	3		
3	S50	travel di	4	1000	0,5	C11	640	sort by AM	DAMIT available	0,00%		10		S50			2,5	5		
4	Dist	km S	res n	fram	ms e	fram	T ex	Name Asteroid	Datum Ereignis	Wahrsch	Distan	M komb	M Aste	M Ste	t max	(R) dr	(V) dr	Schat	Sternl	Geme
326	-1	6	9,4	9,0	56	12,5	40	(204850) 2007 RD121	So 02 Nov, 00:12	0,059	2 km S	9,3	20,1	9,3	0,5	10,2	10,8	6 km	51Å SE	4
365	26	16	7,4	11,4	88	25,0	40	(18340) 1989 OM	Mo 03 Nov, 05:16	0	34 km	9,5	18	9,5	1	7,8	8,5	16 km	41Å W	0
596	0	14	9,1	30,2	56	42,5	40	(15138) 2000 EQ93	So 09 Nov, 00:04	0,36	7 km S	8,7	18,6	8,7	1,7	8,6	9,9	14 km	24Å E	2
619	13	6	5,7	7,9	101	20,0	40	(71631) 2000 EX75	Mo 10 Nov, 21:38	0	16 km S	10	17,2	10	0,8	6,5	7,2	6 km	69Å SE	0
731	29	8	12,0	40,0	20	20,0	40	(130239) 2000 CU39	Sa 15 Nov, 20:26	0	33 km	7,5	20,2	7,5	0,8	12,1	12,7	8 km	73Å S	1
1056	-8,5	19	7,7	23,2	69	40,0	40	(35310) 1997 AX1	Do 27 Nov, 19:22	1	1 km N	9	17,6	9	1,6	7,8	8,6	19 km	28Å NE	3
1097	18	8	5,7	38,0	26	25,0	40	(10562) 1993 UB1	Sa 29 Nov, 02:11	0	22 km	8,7	16	8,7	1	6,6	7,3	8 km	49Å NW	0
1279	0	24	7,4	80,0	20	40,0	40	(15410) 1997 YZ	Fr 05 Dez, 02:18	0,559	12 km E	8,2	17	8,2	1,6	8,4	8,8	24 km	50Å SW	1
1683	5	8	8,4	6,5	78	12,5	40	(196091) 2002 TG103	Do 18 Dez, 22:12	0,225	9 km S	9,4	19	9,4	0,5	8,7	9,6	8 km	41Å SE	0
1805	23	18	10,4	65,0	20	32,5	40	(12690) Kochimiraikagak	Mo 22 Dez, 02:05	0	32 km	6,3	16,2	6,3	1,3	9,4	9,9	18 km	47Å S	0
1884	2,5	5	8,4	3,6	111	10,0	40	(43652) 2002 EQ88	Mi 24 Dez, 04:07	0,002	5 km S	9,4	18,6	9,4	0,4	8,3	9,2	5 km	28Å SW	0
1933	14,5	109	3,6	59,6	106	157,5	40	(676) Melitta	Do 25 Dez, 02:10	0	69 km	9,8	14,6	9,8	6,3	4,1	4,8	109 km	50Å S	18
2076	25	6	7,8	2,6	114	7,5	40	(22085) 2000 AA169	Mo 29 Dez, 22:29	0	28 km	9,5	18,2	9,5	0,3	8,1	8,7	6 km	31Å SE	1
2319	16	8	9,0	2,9	170	12,5	40	(83282) 2001 RD88	So 04 Jan, 03:17	0	20 km	9,9	19,7	9,9	0,5	8,8	9,8	8 km	33Å W	1
2548	0,5	5	8,4	4,2	95	10,0	40	(106810) 2000 XY39	Sa 10 Jan, 17:33	0,437	3 km S	9,2	18,4	9,2	0,4	8,4	9,2	5 km	26Å NE	1
2679	4	6	6,9	35,8	22	20,0	40	(71124) 1999 XR173	Di 13 Jan, 21:52	0,191	7 km N	8,9	17,8	8,9	0,8	7,9	8,9	6 km	72Å S	2
2886	-1,5	5	8,8	5,4	73	10,0	40	(117782) 2005 GL115	Di 20 Jan, 21:57	0,274	1 km E	9,7	20	9,7	0,4	9,5	10,3	5 km	65Å SE	1
3164	27	28	6,9	16,8	155	65,0	40	(3372) Bratijchuk	Fr 30 Jan, 01:08	0	41 km S	9,4	16,7	9,4	2,6	6,6	7,3	28 km	20Å W	0
3284	13	12	11,6	35,0	20	17,5	40	(163407) 2002 RH5	Di 03 Feb, 22:45	0,14	19 km E	7,2	19,2	7,2	0,7	11,4	12	12 km	30Å SE	0
3307	21,5	5	7,8	9,6	52	12,5	40	(88972) 2001 TT58	Mi 04 Feb, 23:15	0	24 km	9,4	18,8	9,4	0,5	8,6	9,4	5 km	62Å SE	0
3364	-2,5	9	10,8	22,5	27	15,0	40	(20710) 1999 XP10	Sa 07 Feb, 00:45	0,997	2 km N	7,8	19,3	7,8	0,6	10,5	11,5	9 km	17Å W	0
3483	27,5	5	8,9	6,6	76	12,5	40	(104924) 2000 JL21	Mo 16 Feb, 21:33	0	30 km	9,9	20,5	9,9	0,5	9,9	10,6	5 km	81Å SE	1
3728	2,5	41	10,5	100,0	20	50,0	40	(5368) Vitagliano	Do 12 Mrz, 21:06	0,066	23 km	7	17,7	7	2	9,8	10,7	41 km	39Å W	2

OW Cloud Filter



JPL page

5368 Vitagliano (1984 SW5)

Classification: Outer Main-belt Asteroid **SPKID:** 20005368 **Related Links:** Ephemeris

Orbit Viewer [show]

Orbit Parameters [show]

Physical Parameters [hide]

Parameter	Value	Units	Sigma	Reference	N
[H] absolute magnitude	11.13			MPO914192	
diameter	34.807	km	0.415	urn:nasa:pds:neowise_diam...	
rotation period	59.36	h		LCDB (Rev. 2023-October);...	[Result based on less than full coverage...
geometric albedo	0.058		0.011	urn:nasa:pds:neowise_diam...	

[Result based on less than full coverage, so that the period may be wrong by 30 percent or so.] REFERENCE LIST:[Warner, B.D.; Stephens, R.D.; Coley, D.R. (2017) Minor Planet Bull. 44, 130-137.]

https://ssd.jpl.nasa.gov/horizons

(5368) Vitagliano – DAMIT (Uni Prag)

Database of Asteroid Models from Inversion Techniques

damit.cuni.cz/projects/damit/?q=5368

DAMIT Models Tumblers Tables Export Documentation More

Modell 13151 vs. 13152:

Devation at 12.03.2026: 0,33 revoultions (118,6°)

P: ± 0,001h equals ± 0,11 revoultions(±39,5°)

5368

(10757 asteroid

Models

	λ	β	P	Files
(5368) Vitagliano				LCref | lc.txt | lc.jsc
View model	35	62	30.281	IAUspin | shape.pr
View model	205	63	30.278	IAUspin | shape.pr

vgl. JPL: 59,36h, DAMIT 29,69h: factor 2 with Δ -2%

https://damit.cuni.cz **Ďurech** & Hanuš (2023)

damit.cuni.cz/projects/damit/asteroid_models/view/13151

DAMIT Models Tumblers Tables Export Documentation More

(5368) Vitagliano – Model 13151

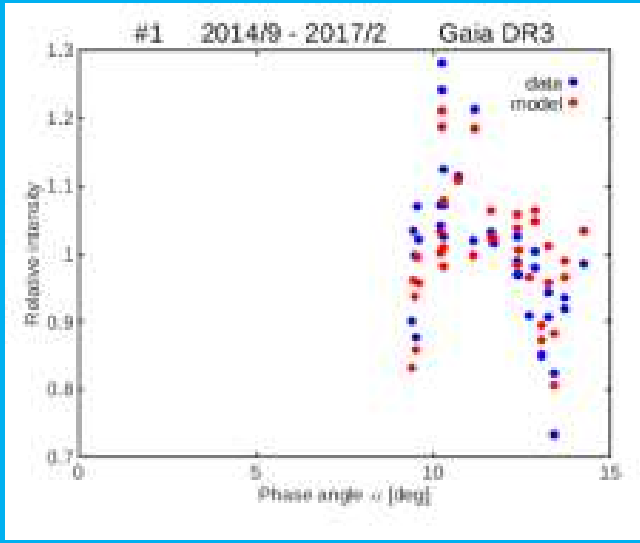
Details

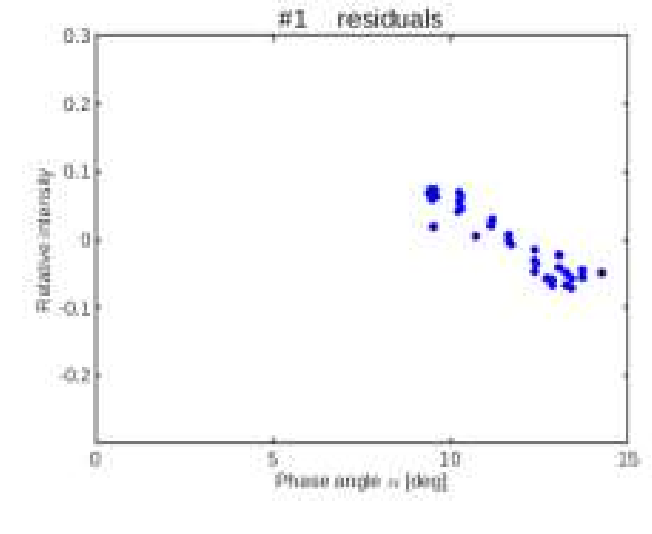
ID	13151
Asteroid	(5368) Vitagliano
λ	35
β	62
P	30.281
YORP	
t_0	2456916
φ_0	0
Light Scattering Model	LSL
p_1	0.1
p_2	0.5
p_3	0.1
p_4	-0.68

15.09.2014

(5368) Vitagliano – ISAM

Infrared Standard Asteroid Model






```
Satellite IR diameters for asteroid (5368)

  NEOWISE:  34.8 ± 3.2km:  albedo 0.058
     AcuA:  39.3 ± 7.6km:  albedo 0.046

Weighted diameter:  35.5 ±  3.2 km
[including recommended increase in uncertainty]

Individual diameter and albedo measurements
   34.8 ±  3.2km:  albedo 0.058:  NEOWISE
   39.3 ±  8.4km:  albedo 0.046:  AcuA

Estimated diameter, absolute mag. H0 = 11.17
   albedo 0.40 => diameter = 12 km
   albedo 0.35 => diameter = 13 km
   albedo 0.30 => diameter = 14 km
   albedo 0.25 => diameter = 16 km
   albedo 0.20 => diameter = 17 km
 * albedo 0.15 => diameter = 20 km *
   albedo 0.10 => diameter = 25 km
   albedo 0.09 => diameter = 26 km
   albedo 0.08 => diameter = 27 km
   albedo 0.07 => diameter = 29 km
   albedo 0.06 => diameter = 32 km
   albedo 0.05 => diameter = 35 km
   albedo 0.04 => diameter = 39 km
   albedo 0.03 => diameter = 45 km
   albedo 0.02 => diameter = 55 km
   albedo 0.01 => diameter = 78 km
  Trojans 0.06 => diameter = 32 km
```

Light curve fits for (5368) model 13152

Event JD = 2461112.3375

Daily rotation rate = 285.355038°/day (= model 13152)

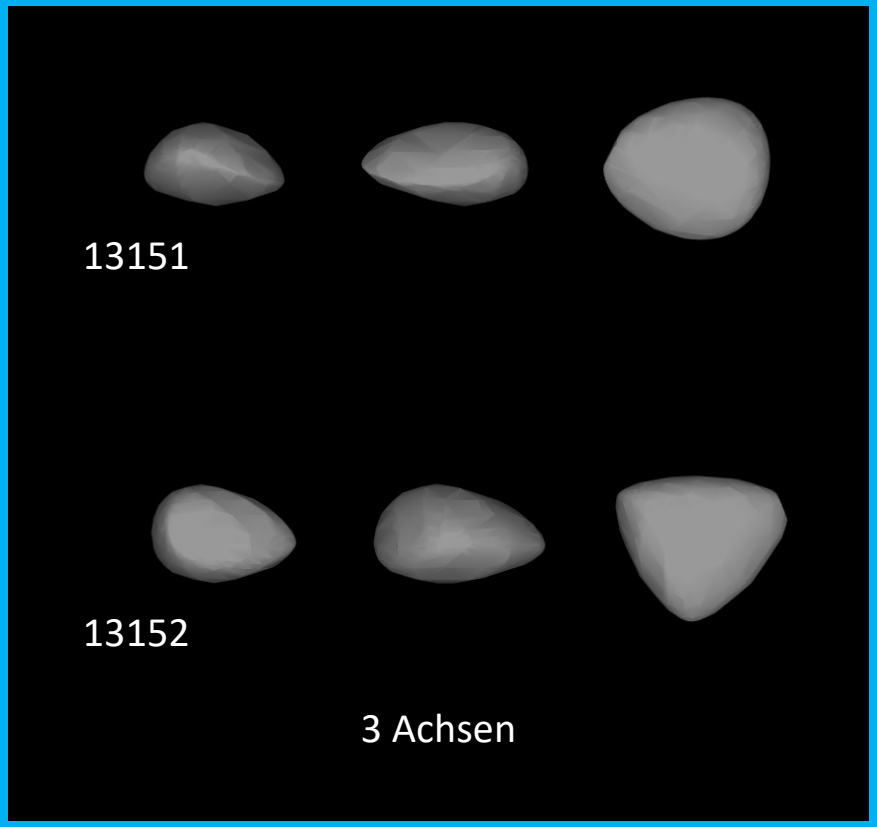
note: 0,000001°/day equals ± 1,3 sec.

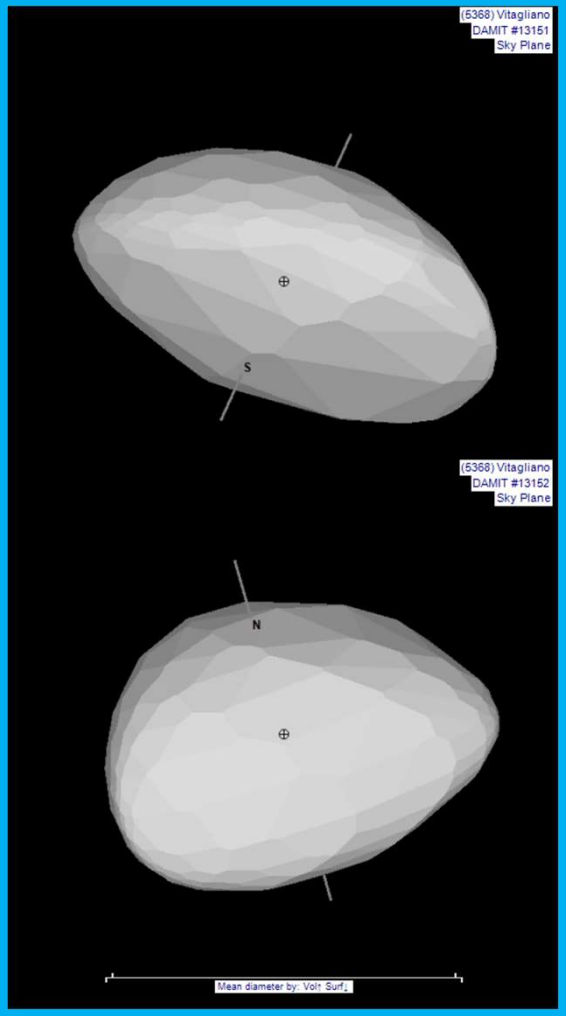
Occult v4.2025.9.10

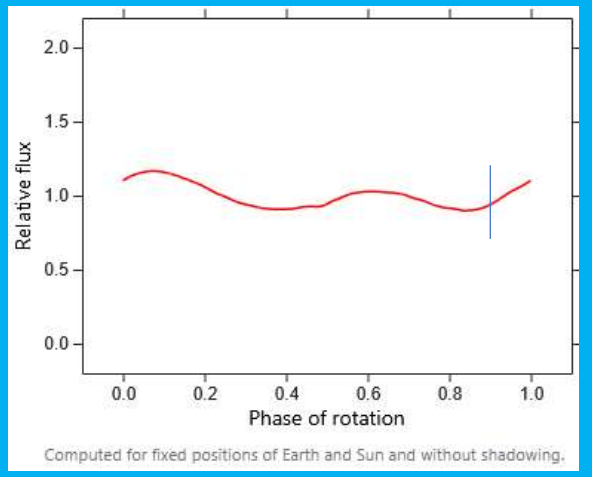
NEOWISE: 2009-2013/2024 (more precisely)

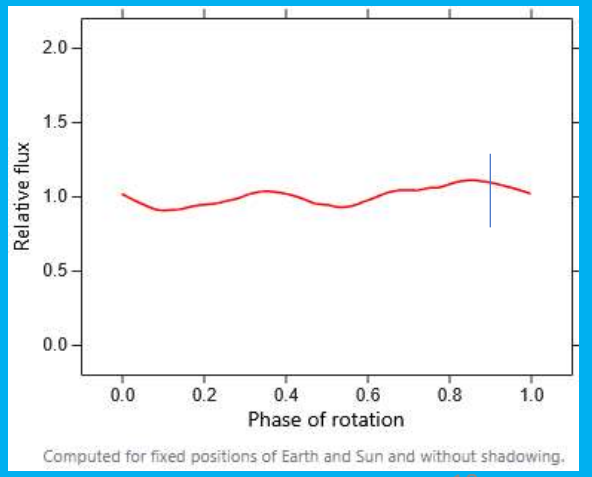
AcuA = Asteroid Catalog using AKARI: 2006 - 2011 (addendum)

(5368) Vitagliano – DAMIT model 13151, 13152 (occult4)

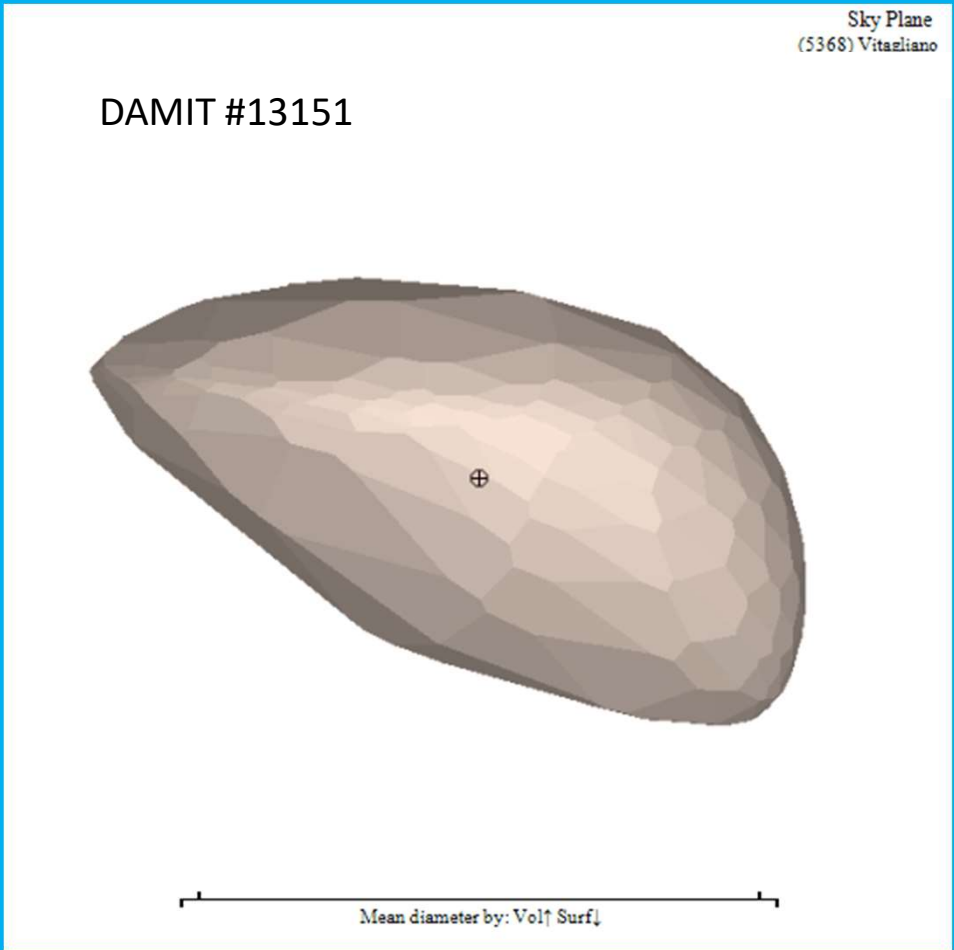


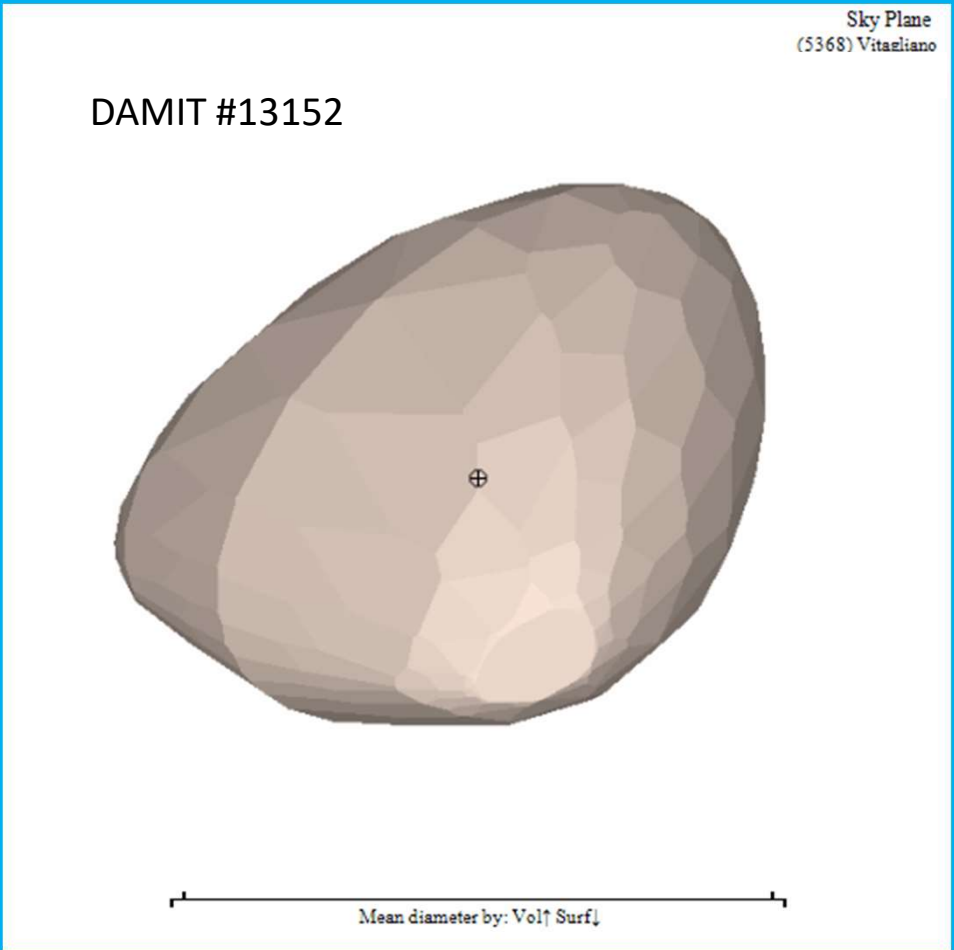




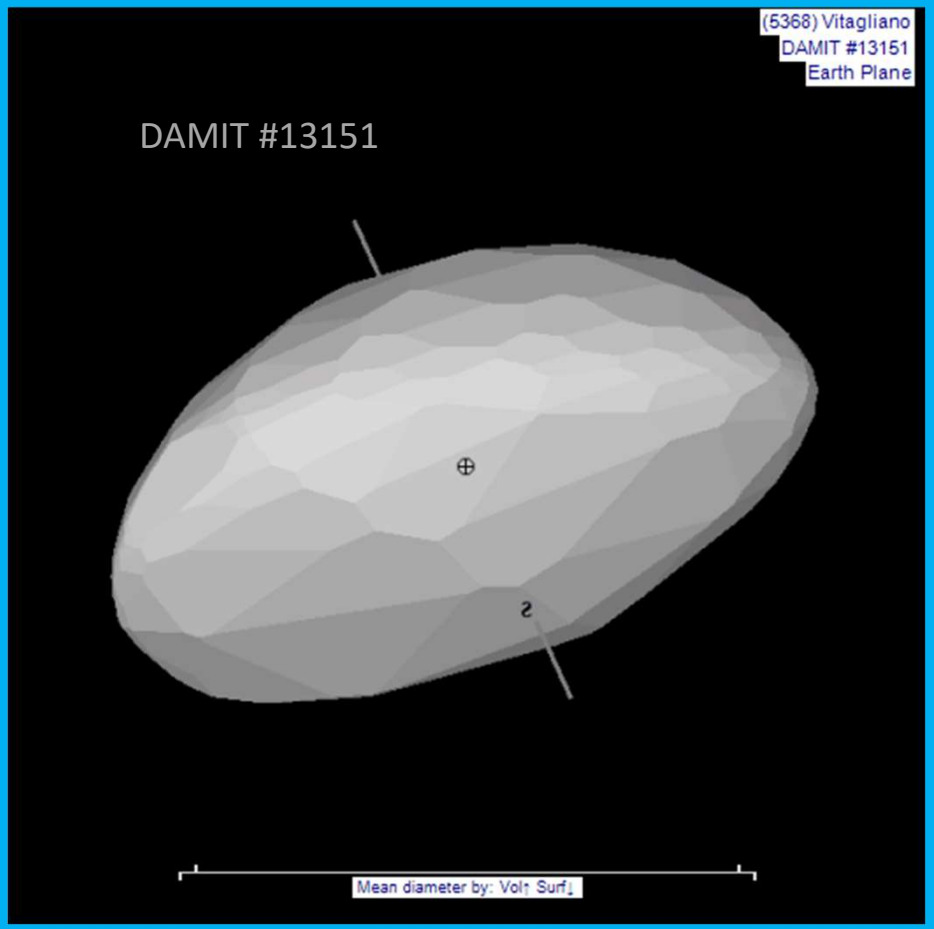


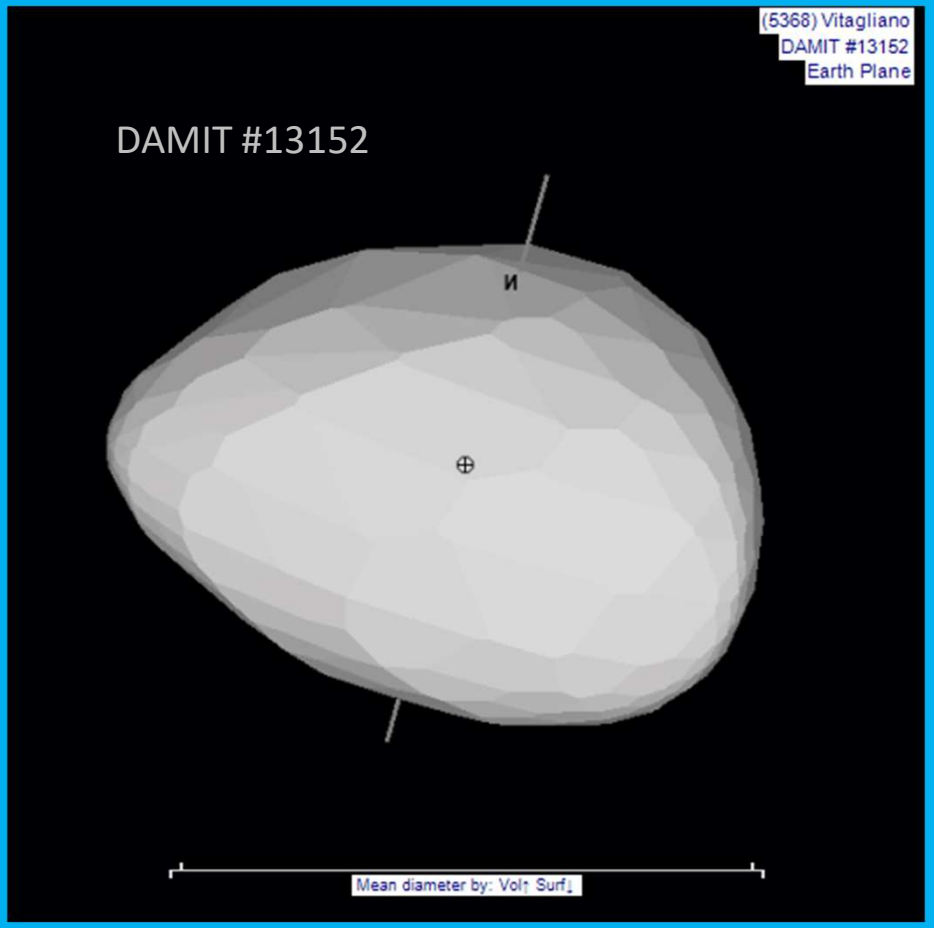
(5368) Vitagliano – animated 30h



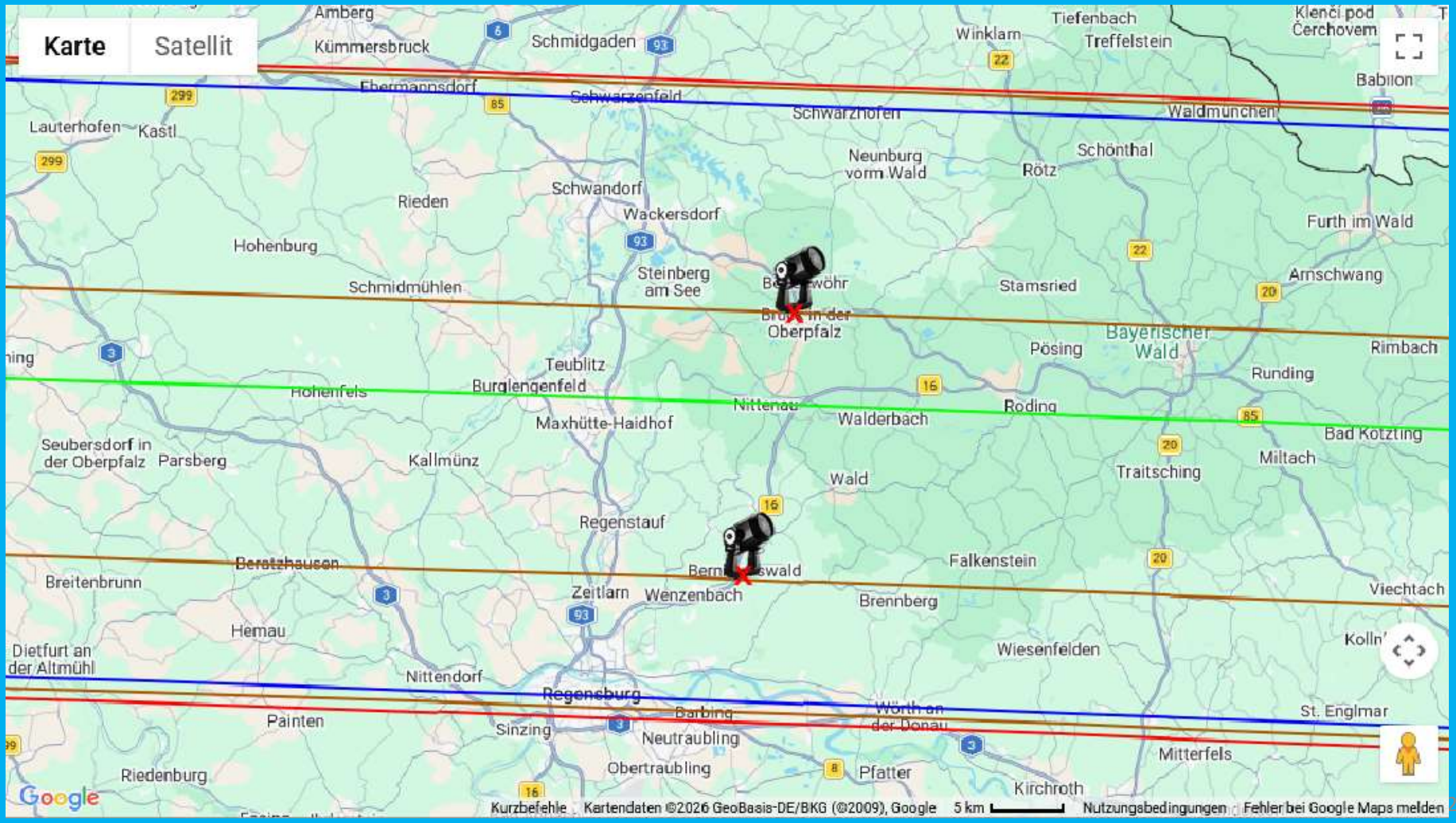


(5368) Vitagliano – animated ±5° / ±25 min.

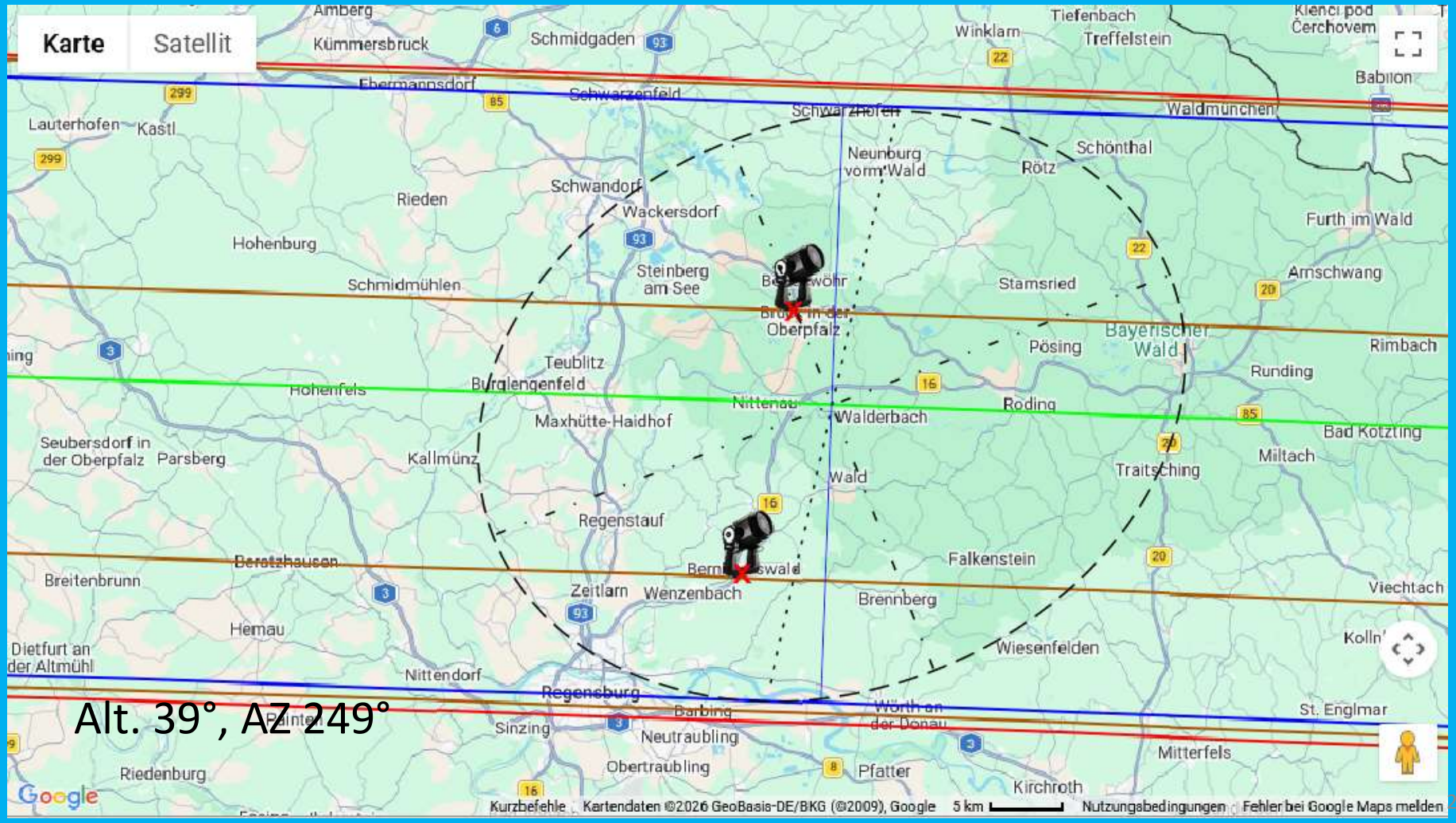




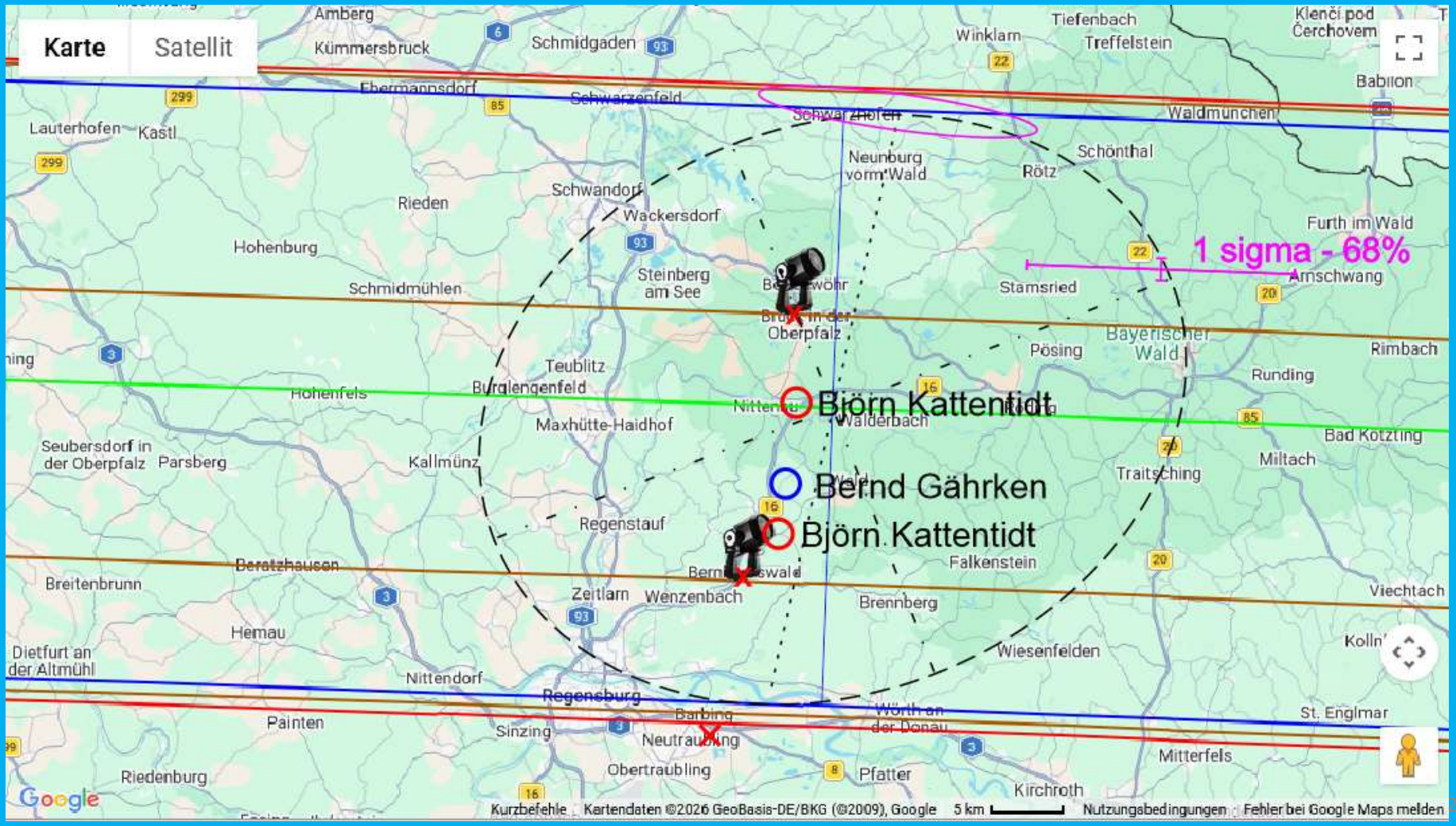
Expedition: Star occultation path and site planning



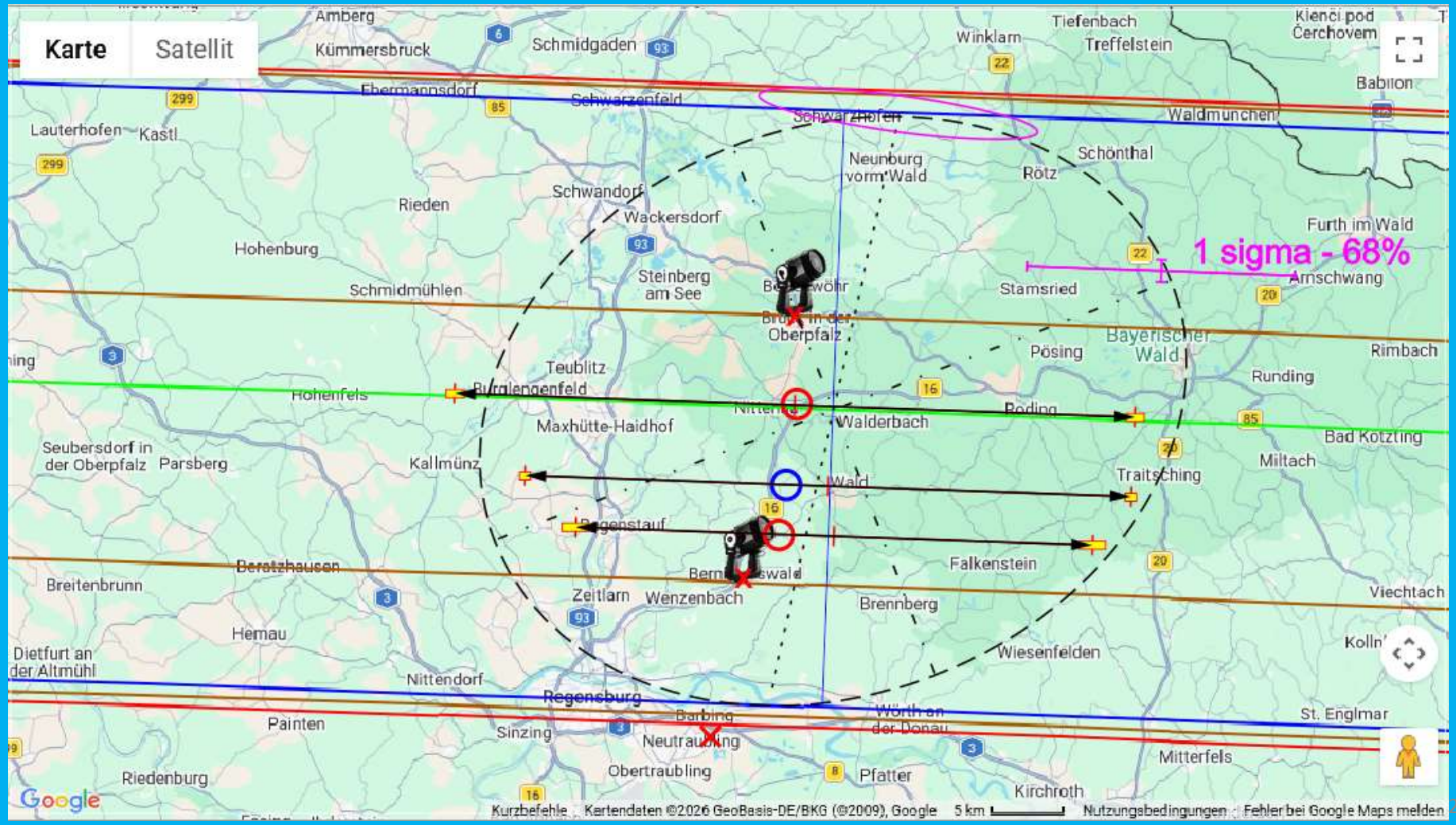
Projection as a spherical asteroid



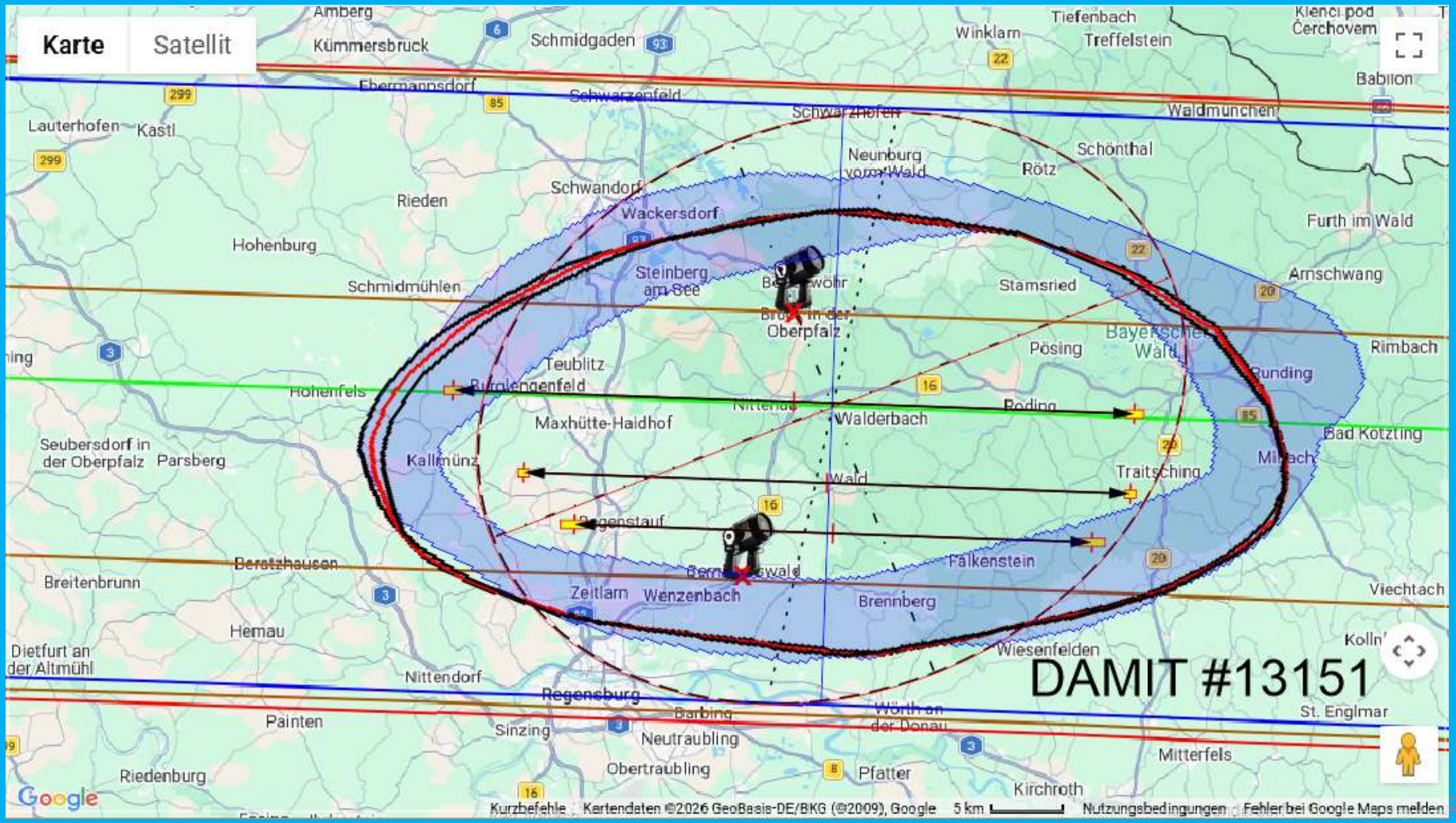
Actual observation locations



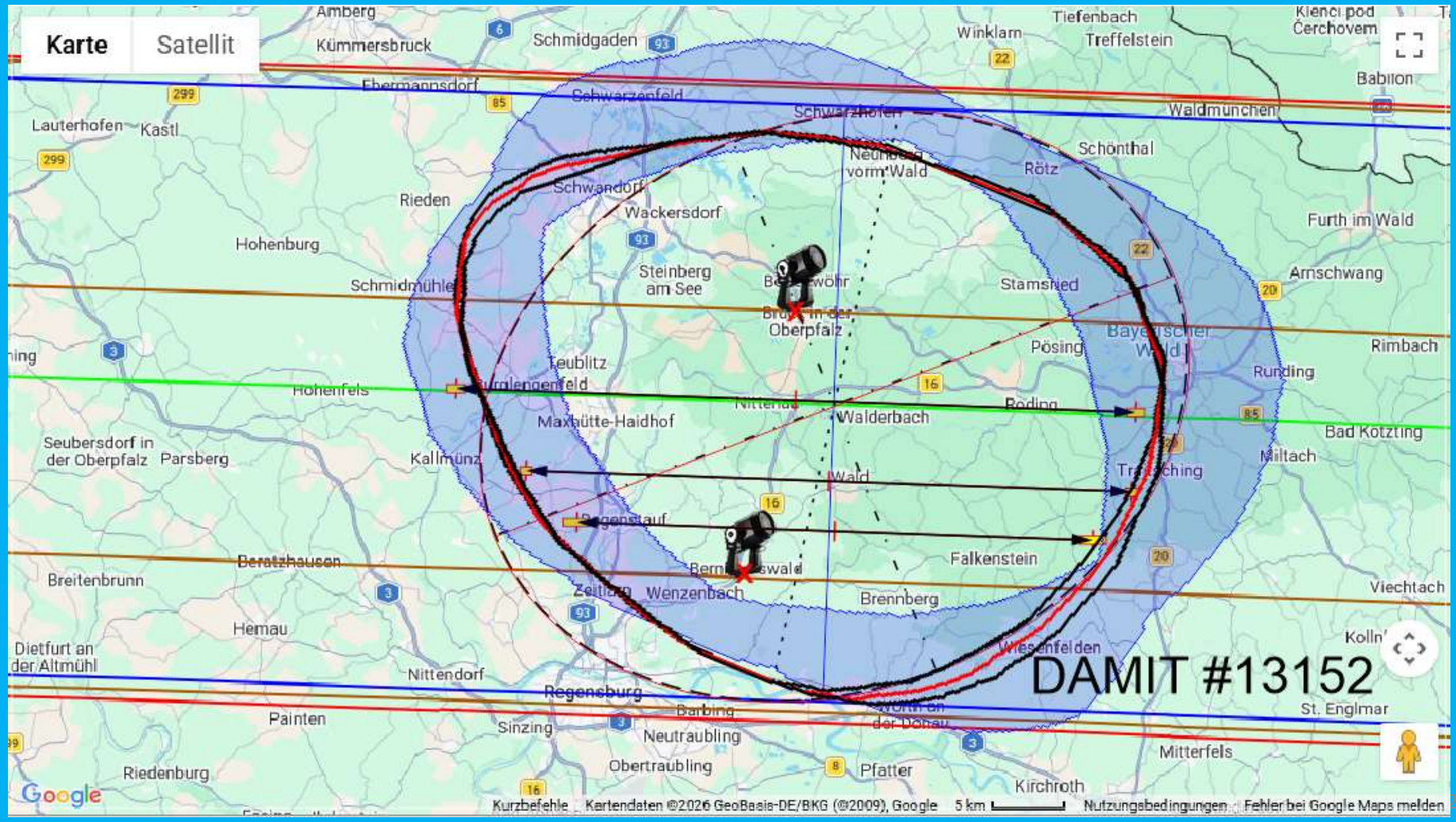
Measured times, projected and shifted



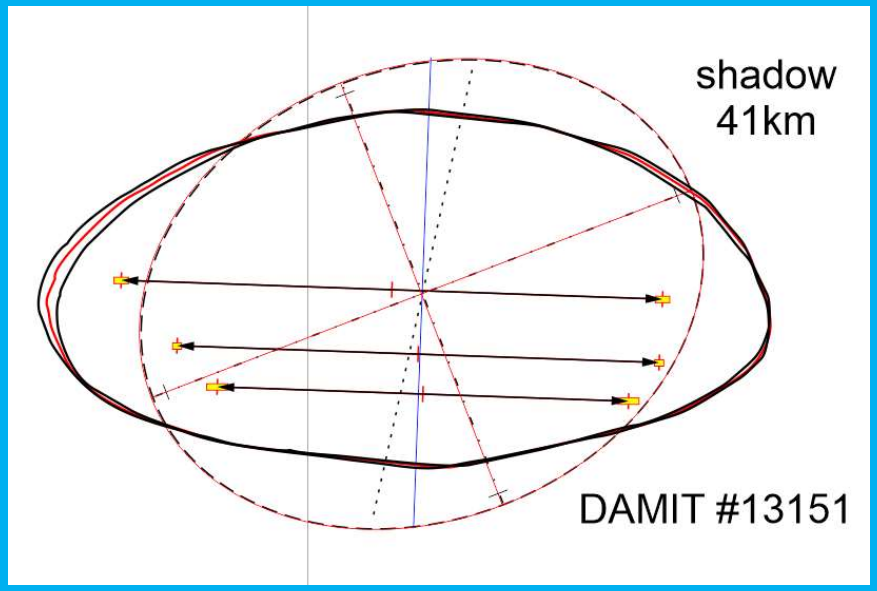
(5368) Vitagliano – model #1

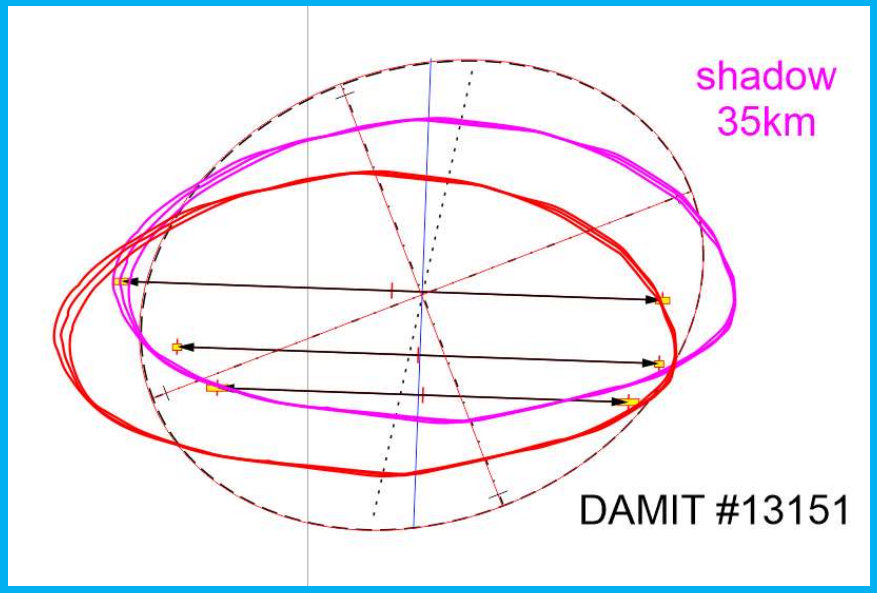


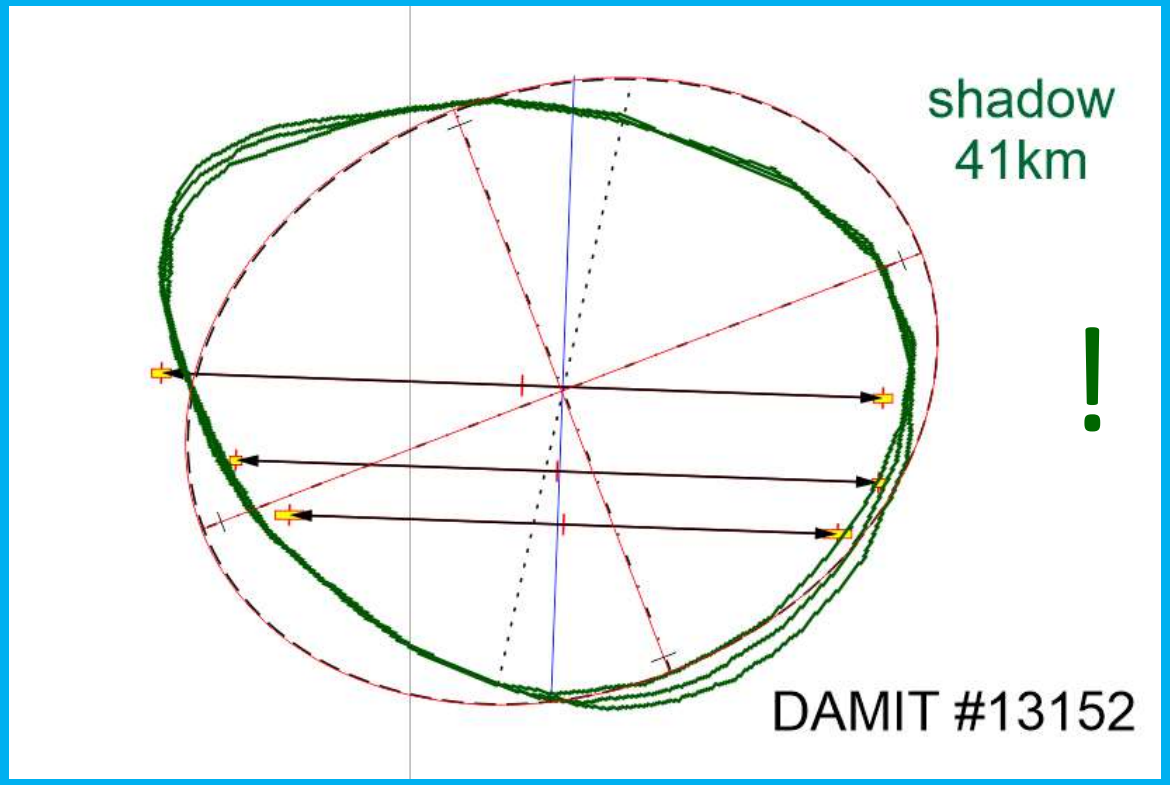
(5368) Vitagliano – model #2



Result


shadow
41km
DAMIT #13151


shadow
35km
DAMIT #13151


shadow
41km
!
DAMIT #13152

(5368) Vitagliano – model #2

```
Observed chord lengths

Length       # Observer

5368 Vitagliano       2020 Apr  2       UCAC4 447-054339
  13.9 km    1 D Gault

5368 Vitagliano       2022 Oct  9       UCAC4 360-195775
  37.9 km    1 A Howell

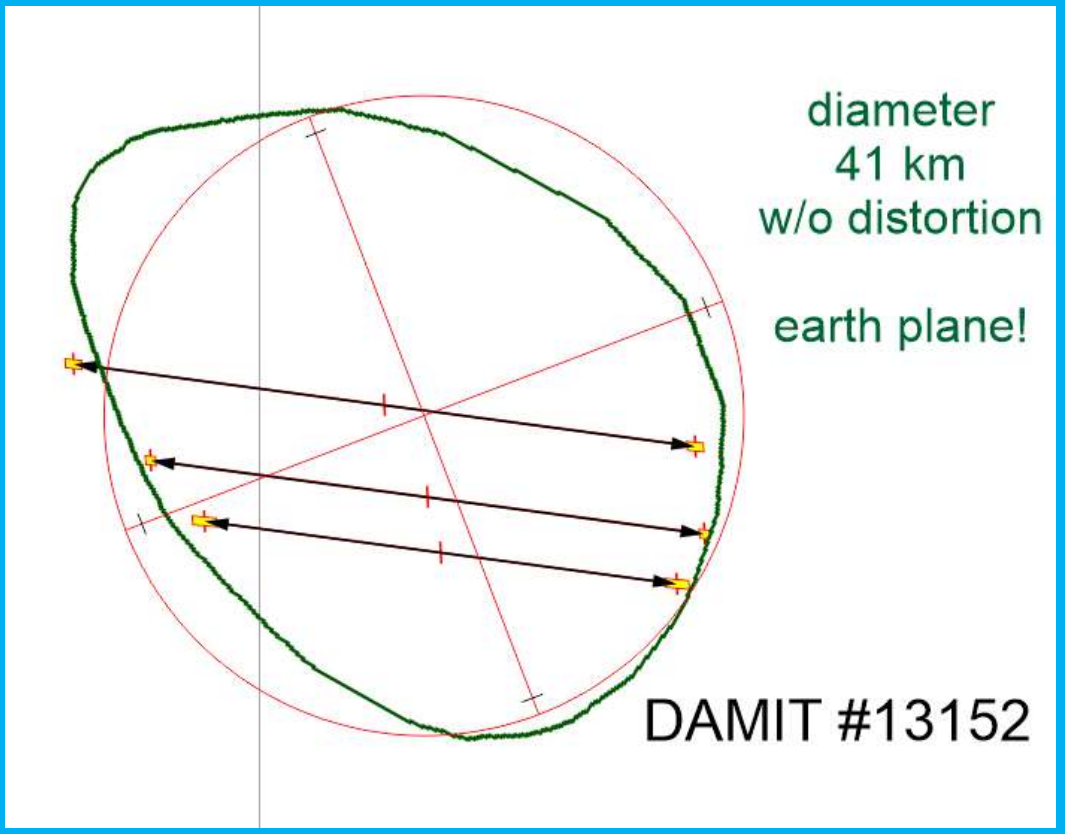
5368 Vitagliano       2023 Sep 21       UCAC4 401-130857
  33.8 km    1 P Maley

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Satellite IR diameters for asteroid (5368)

   NEOWISE:   34.8 ± 3.2km:   albedo 0.058
      AcuA:   39.3 ± 7.6km:   albedo 0.046

Weighted diameter:  35.5 ±  3.2 km
[including recommended increase in uncertainty]
________________________________________________
Individual diameter and albedo measurements
    34.8 ±  3.2km:   albedo 0.058:   NEOWISE
    39.3 ±  8.4km:   albedo 0.046:   AcuA
```

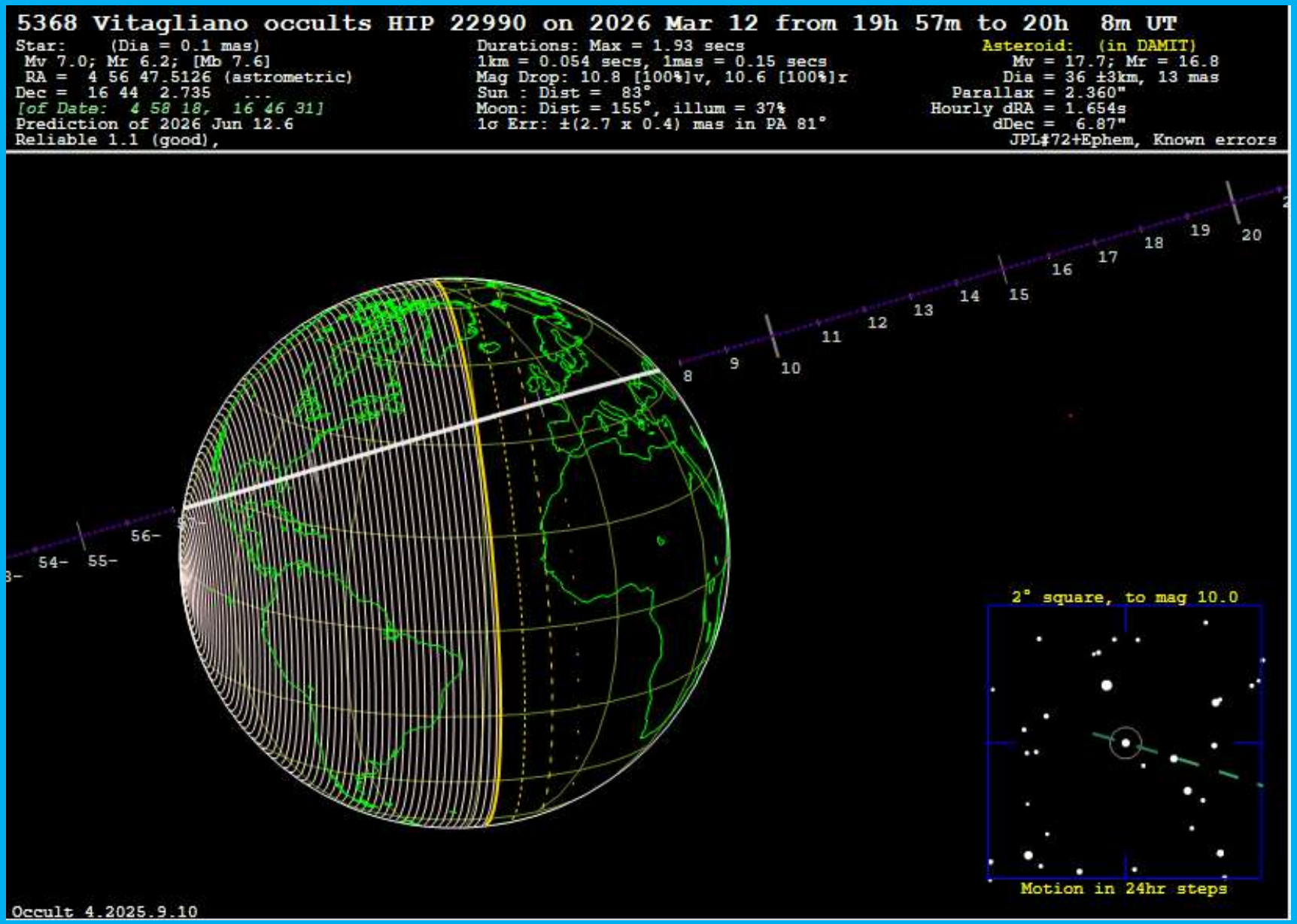


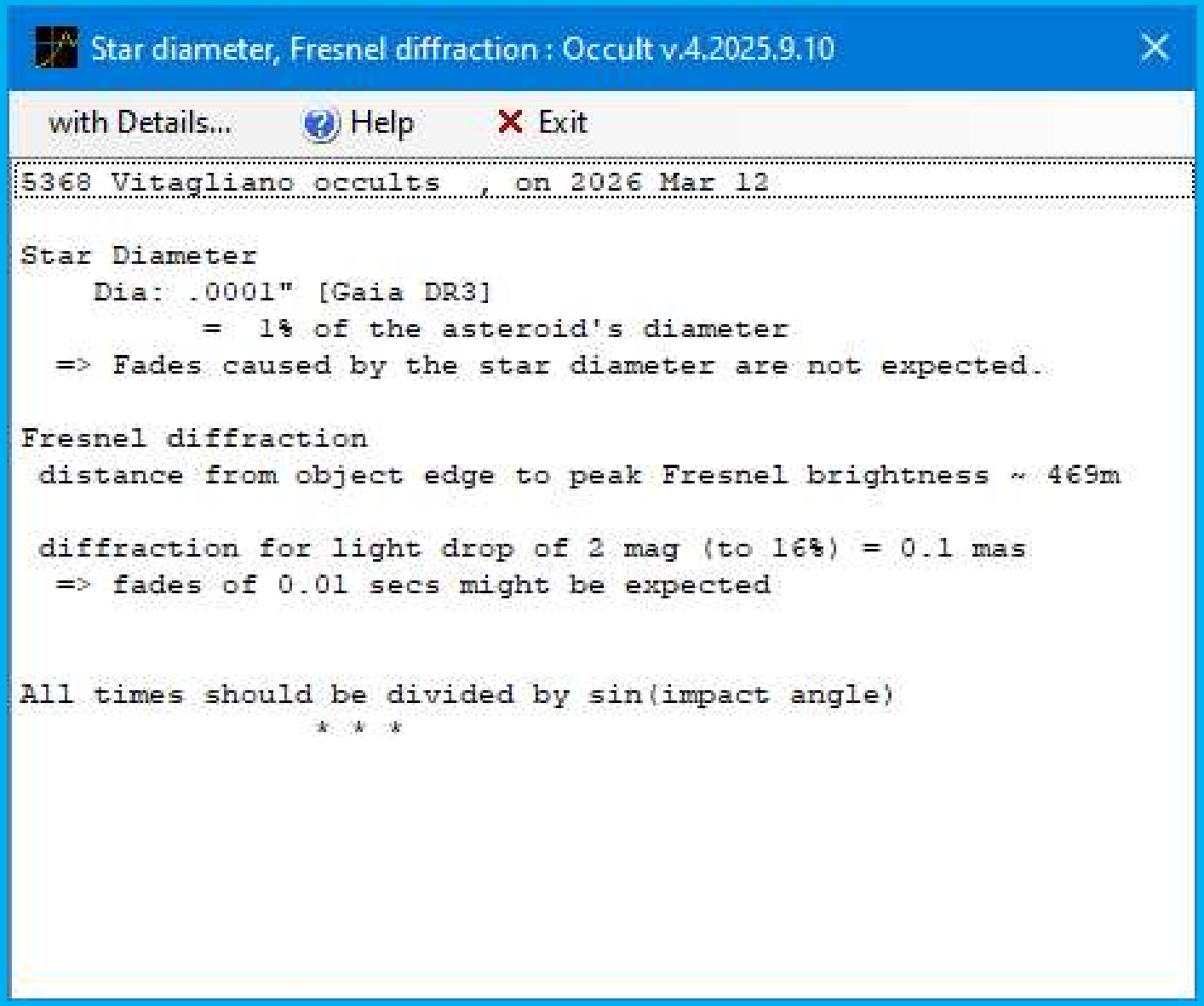
Occult v4.2025.9.10

(5368) Vitagliano – „models“
scale 1 : 1000 000






5368 Vitagliano occults HIP 22990 on 2026 Mar 12 from 19h 57m to 20h 8m UT
Star: (Dia = 0.1 mas)
Mv 7.0; Mr 6.2; [Mb 7.6]
RA = 4 56 47.5126 (astrometric)
Dec = 16 44 2.735 ...
[of Date: 4 58 18, 16 46 31]
Prediction of 2026 Jun 12.6
Reliable 1.1 (good),
Durations: Max = 1.93 secs
1km = 0.054 secs, 1mas = 0.15 secs
Mag Drop: 10.8 [100%]v, 10.6 [100%]r
Sun : Dist = 83°
Moon: Dist = 155°, illum = 37%
1σ Err: ±(2.7 x 0.4) mas in PA 81°
Asteroid: (in DAMIT)
Mv = 17.7; Mr = 16.8
Dia = 36 ±3km, 13 mas
Parallax = 2.360"
Hourly dRA = 1.654s
dDec = 6.87"
JPL#72+Ephem, Known errors
8
9
10
11
12
13
14
15
16
17
18
19
20
54-
55-
56-
2° square, to mag 10.0
Motion in 24hr steps
Occult 4.2025.9.10


Star diameter, Fresnel diffraction : Occult v.4.2025.9.10
with Details...
Help
Exit
5368 Vitagliano occults , on 2026 Mar 12
Star Diameter
Dia: .0001" [Gaia DR3]
= 1% of the asteroid's diameter
=> Fades caused by the star diameter are not expected.
Fresnel diffraction
distance from object edge to peak Fresnel brightness ~ 469m
diffraction for light drop of 2 mag (to 16%) = 0.1 mas
=> fades of 0.01 secs might be expected
All times should be divided by sin(impact angle)
* * *